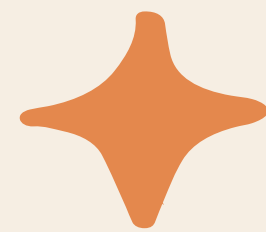


GENERAL MATHEMATICS

GRADE 11 – TERM 1 WEEK 9 & 10



LEARNING COMPETENCIES



Classify functions based on their graphs and/or equations to determine if they are linear, absolute value, quadratic, square root, or cube root.

LEARNING COMPETENCIES



- ✦ Sketch the graph of the following functions: linear, absolute value, quadratic, square root, or cube root with or without technology.
- ✦ Apply the functions to model and solve real-life problems.

Activity A.1.1:

PRETEST on FUNCTIONS

Instructions: Choose the letter of the correct answer.

1. Which of the following best defines a function?
- A. A relation where each input has exactly one output
 - B. A relation where inputs can have multiple outputs
 - C. A set of ordered pairs
 - D. A graph that forms a straight line

Answer: A

2. What is the slope of the linear function
 $f(x) = 5x - 2$?

- A. -2
- B. 5
- C. 2
- D. -5

Answer: B

3. Which of the following represents a quadratic function?

A. $y = 3x + 5$

B. $y = 3x^2 - 2x + 1$

C. $y = 1/x$

D. $y = x^3$

Answer: B

4. What is the shape of the graph of a quadratic function?

- A. Line
- B. Circle
- C. Parabola
- D. Triangle

Answer: C

5. What is the y-intercept of the line $y = -2x + 5$?

A. -2

B. 5

C. -5

D. 2

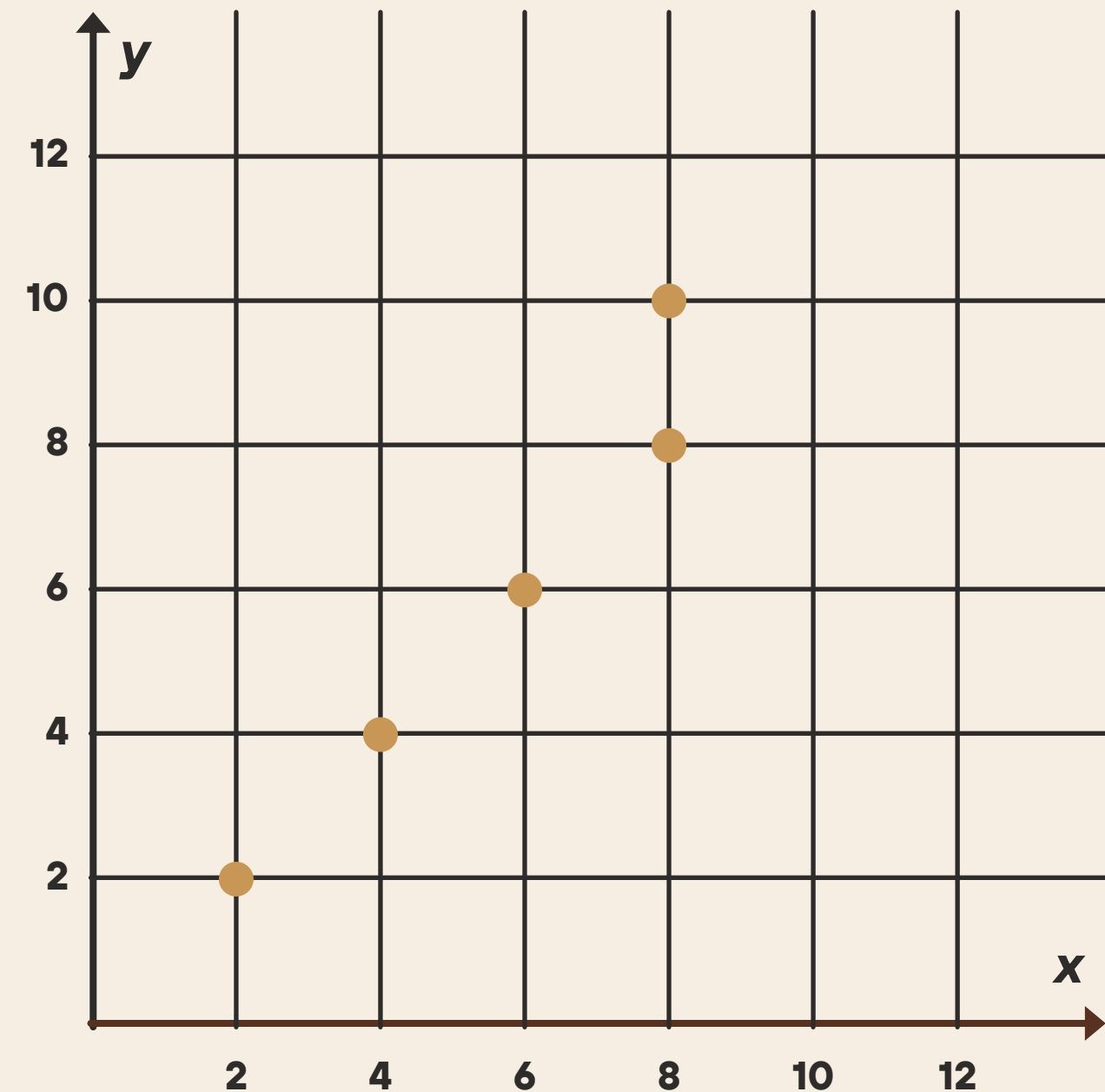
Answer: B

LET'S RECALL AND ANALYSE!

Do you still remember how to fill in the
table of values based on a given graph?

Complete the table of values below.

x					
y					

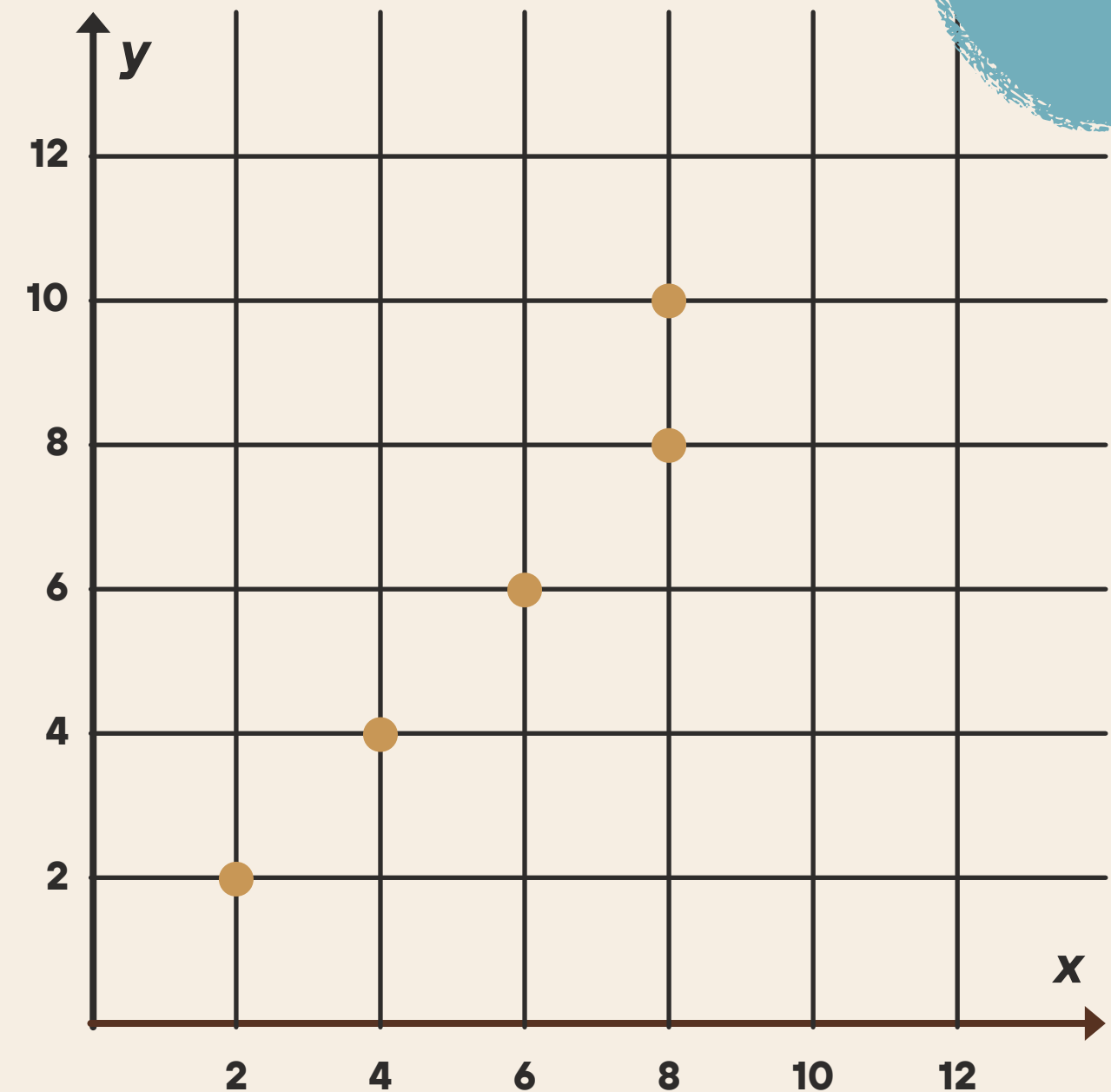


LET'S RECALL AND ANALYSE!

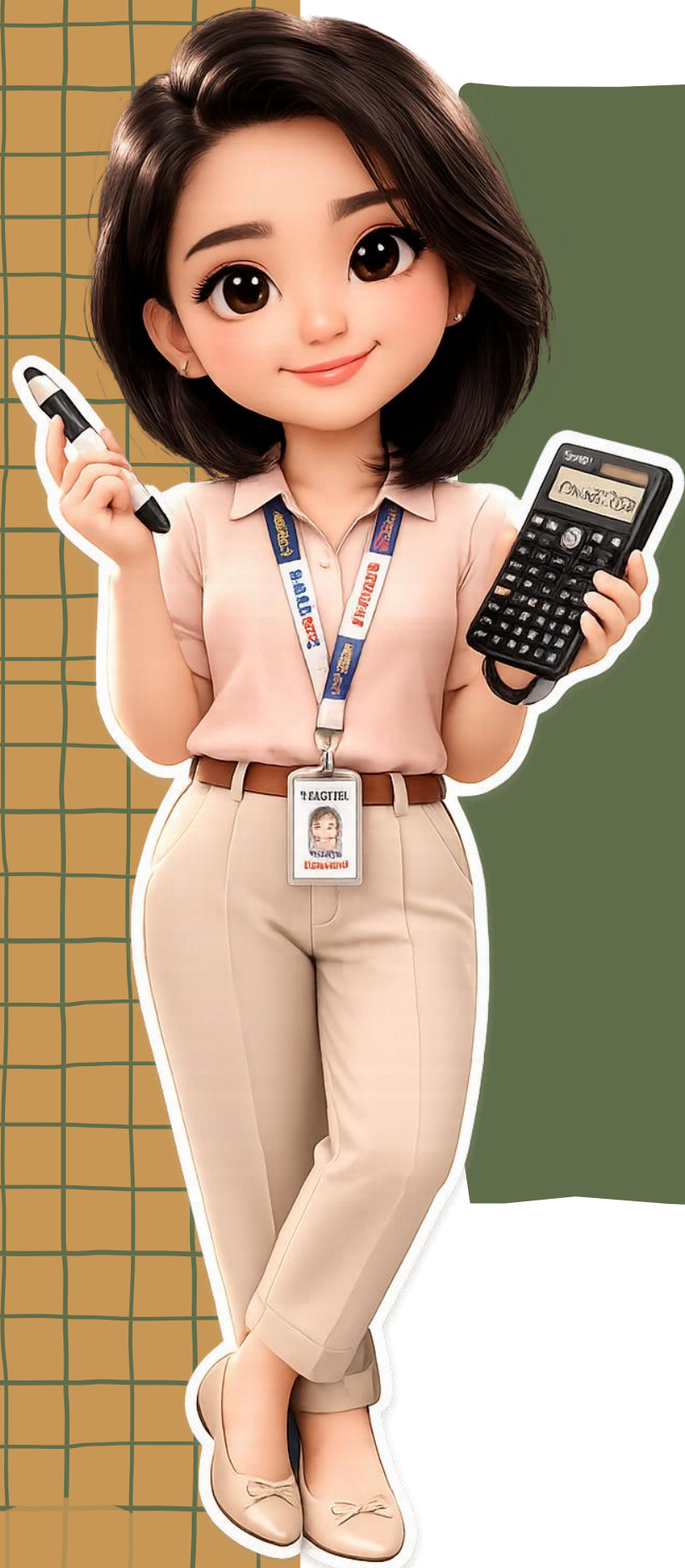
Do you still remember how to fill in the
table of values based on a given graph?

Complete the table of values below.

x	2	4	6	8	8
y	2	4	6	8	10



ANSWER
KEY



WHAT IS A FUNCTION?



- A function is a special type of relation where each **input only has one output**.
- Functions are a way of connecting **input** values to their corresponding **output** values.
- The relation is a function if, for every x -value, there is exactly **one y -value**.

WHAT IS A FUNCTION?

Functions are a way of connecting **input** values to their corresponding **output** values.

Example:

Ordering a coffee for ₹50

Pay ₹50, and you get 1 coffee.

Pay ₹100, and you'll get 2 coffees, and so on.

The input affects the output in a predictable way.



GRAPHING A FUNCTION USING THE TABLE OF VALUES

Given the equation **$y = 2x$** .

Step 1:

Construct the table of values.

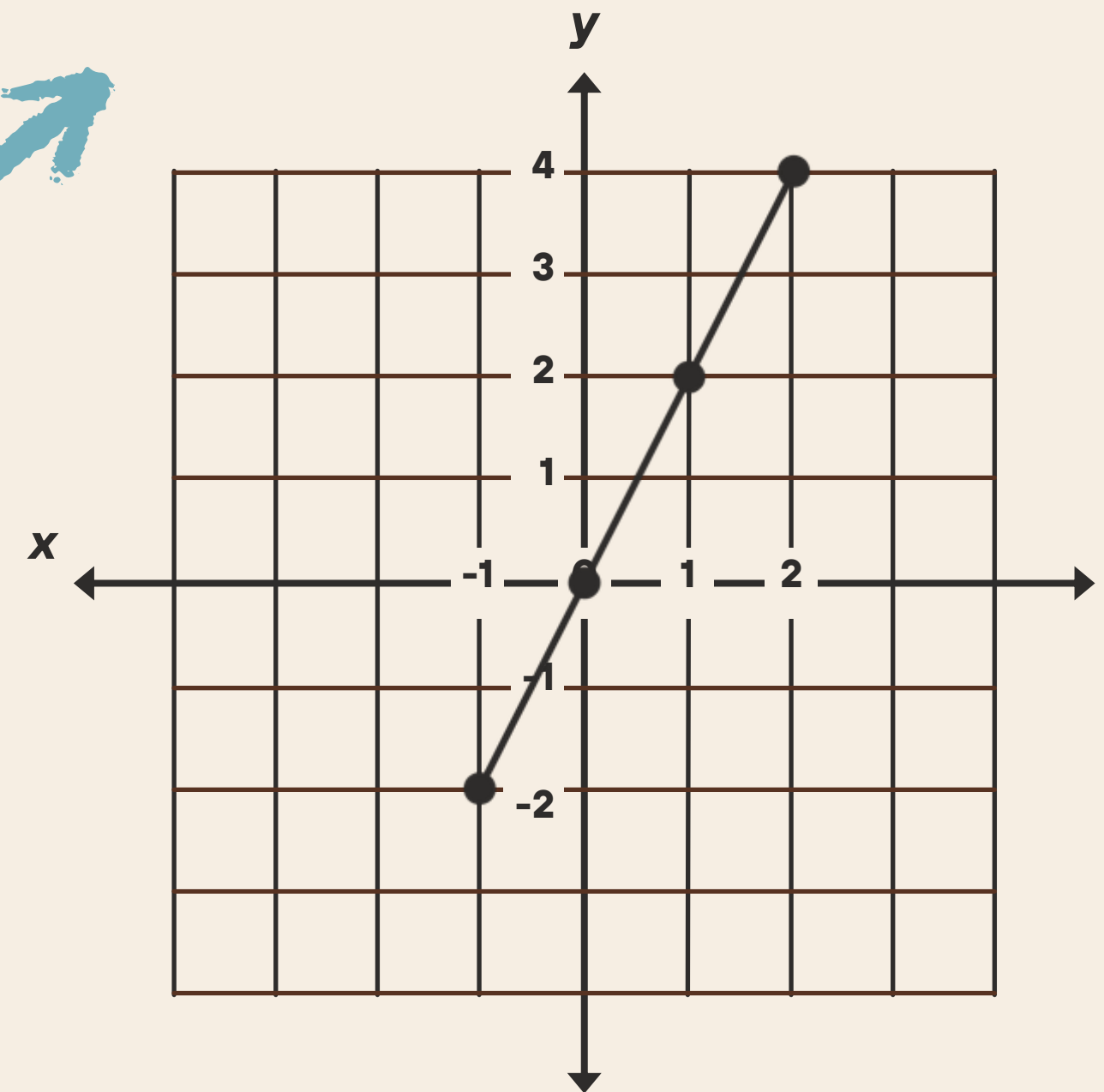
Do this by substituting values of x to $y=2x$.

x	-1	0	1	2
y	-2	0	2	4

Notice that for every value of x , there is only one value of y .

Step 2:

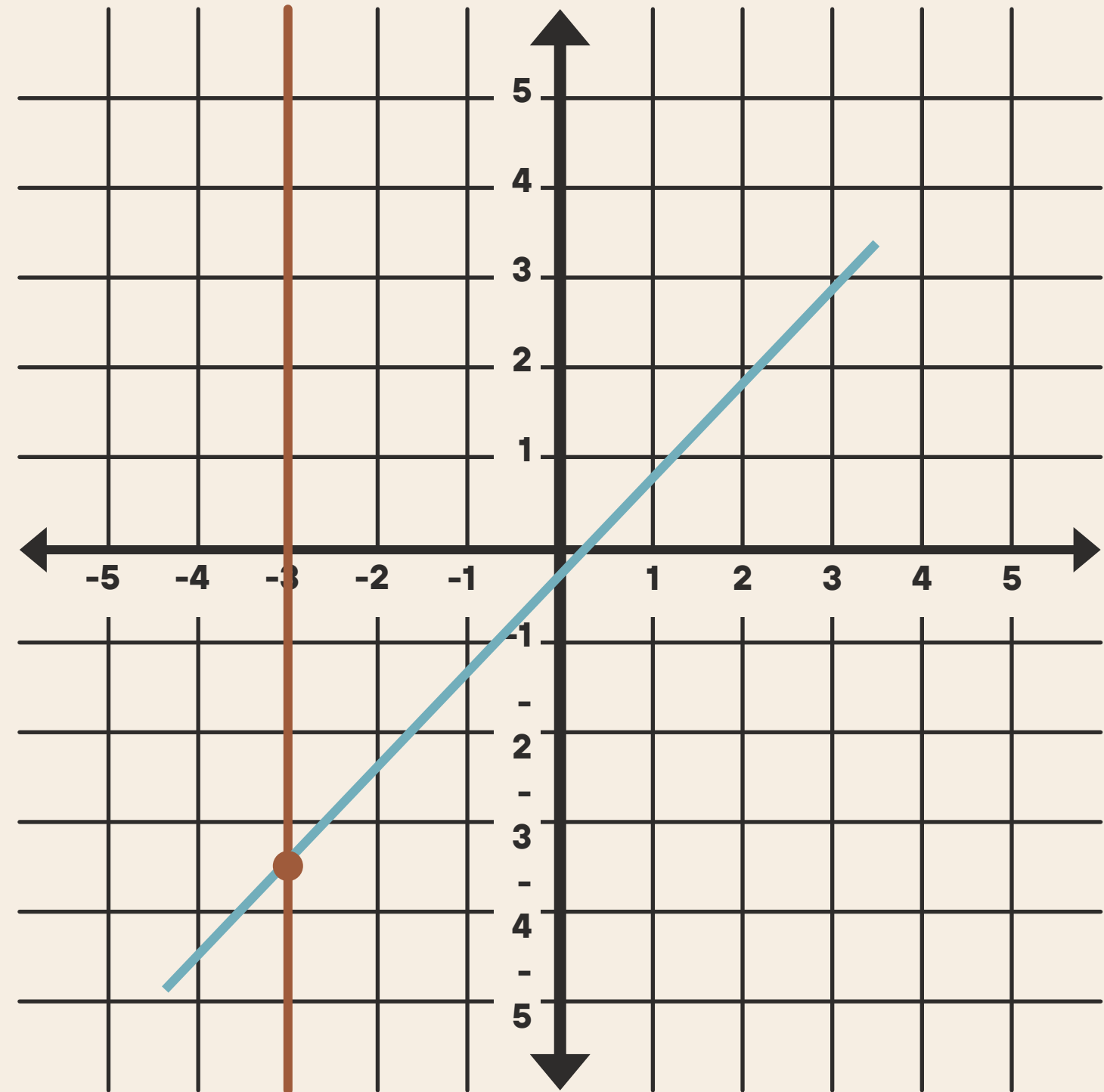
Plot the points.



VERTICAL LINE TEST

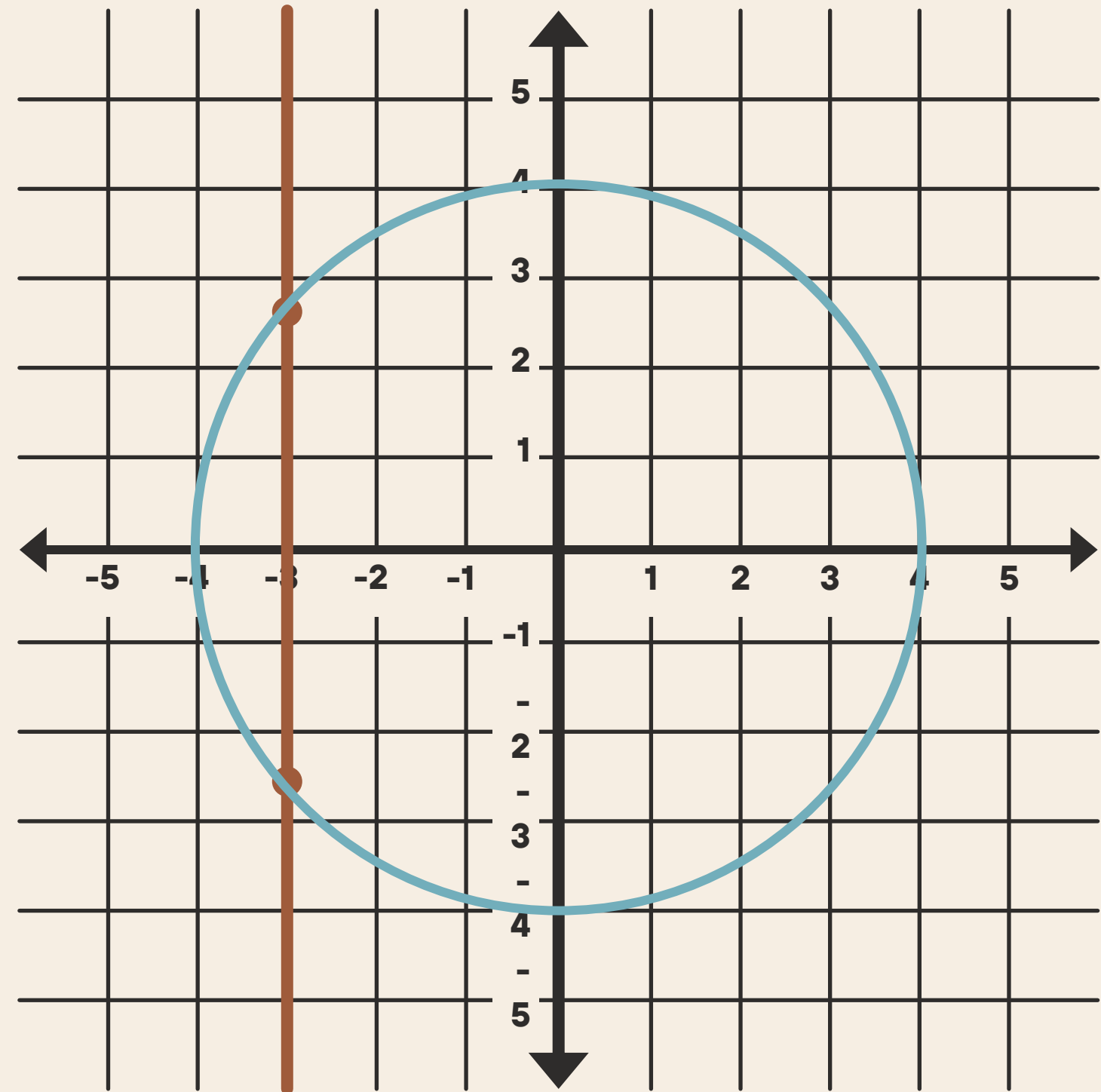
Use the vertical line test to check whether a graph of a relation is a function.

If a vertical line, drawn anywhere on the graph, can touch the graph only once, then it is a **function**.



VERTICAL LINE TEST

If a vertical line, drawn anywhere on the graph, can touch the graph more than once, then the relation is **not a function**.



Shortcut for Graphing Linear Functions

1. Start with the y -intercept: • The equation is in slope-intercept form:

$$y = mx + b$$

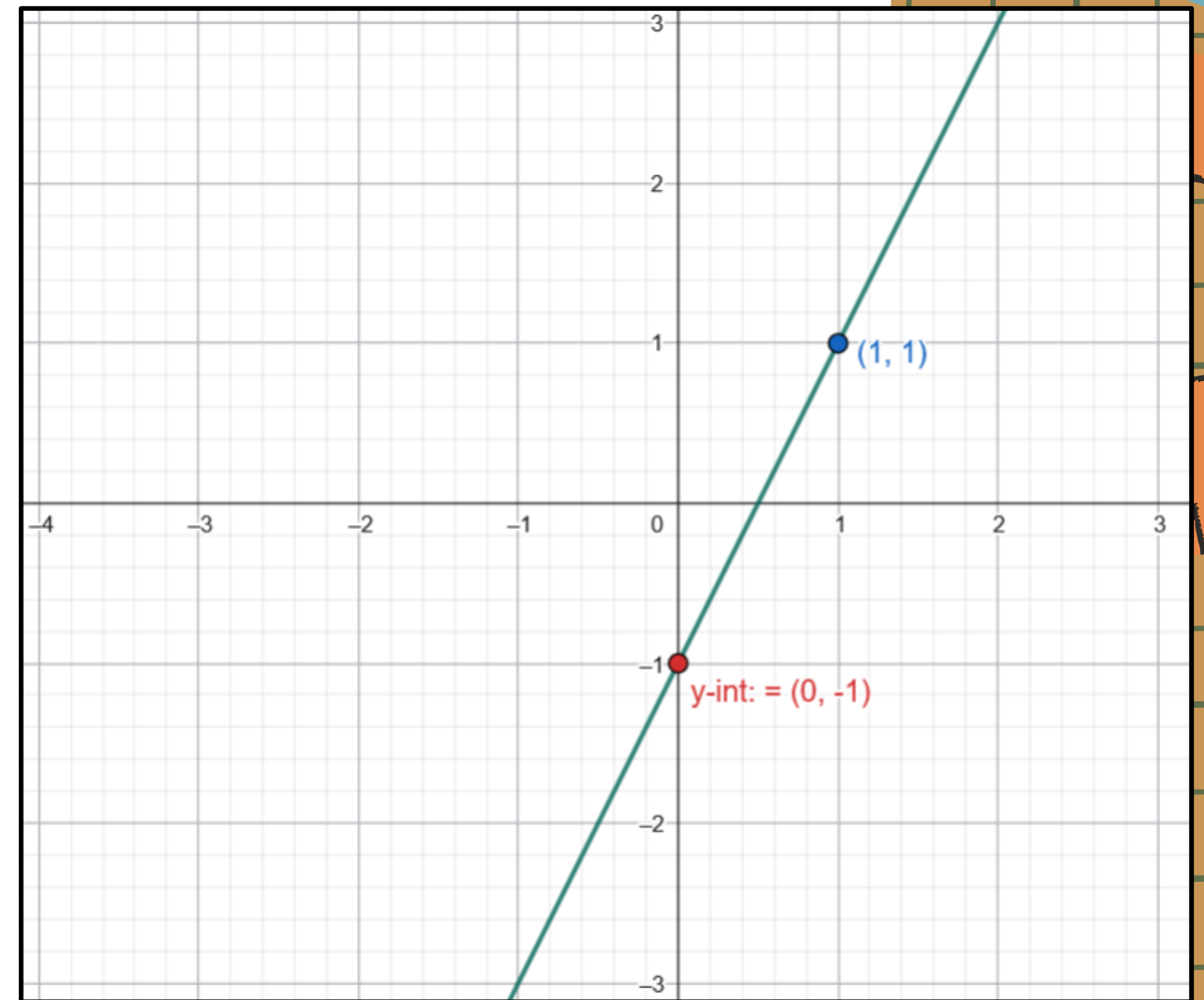
• Here, $b = -1$, so plot the point $(0, -1)$ on the y -axis.

2. Use the slope: • The slope $m = 2$ means **rise 2, run 1**.

• From $(0, -1)$, move **up 2 units** and **right 1 unit** to plot the next point: $(1, 1)$.

3. Draw the line through the points.

Here is the graph of the function
 $y = 2x - 1$



Shortcut for Graphing $y=x^2+2$

1. Identify the vertex:

- The function is in the form $y=x^2+c$, so the vertex is at $(0, 2)$.

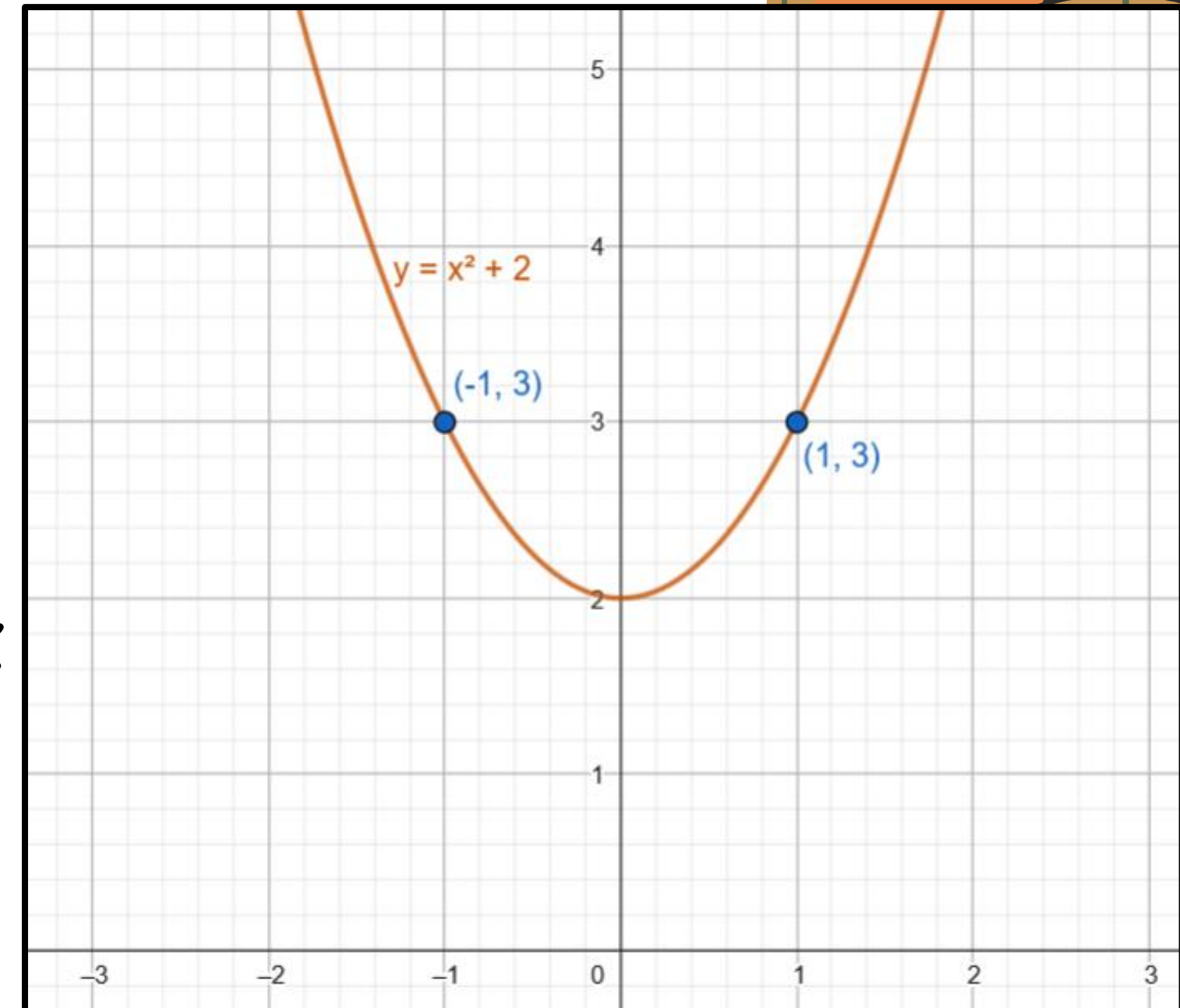
2. Plot symmetric points:

- Choose values of x around the vertex (e.g., $x=-1$ and $x=1$):
- $y=(-1)^2+2=3$
- $y=(1)^2+2=3$
- Plot the points $(-1, 3)$ and $(1, 3)$.

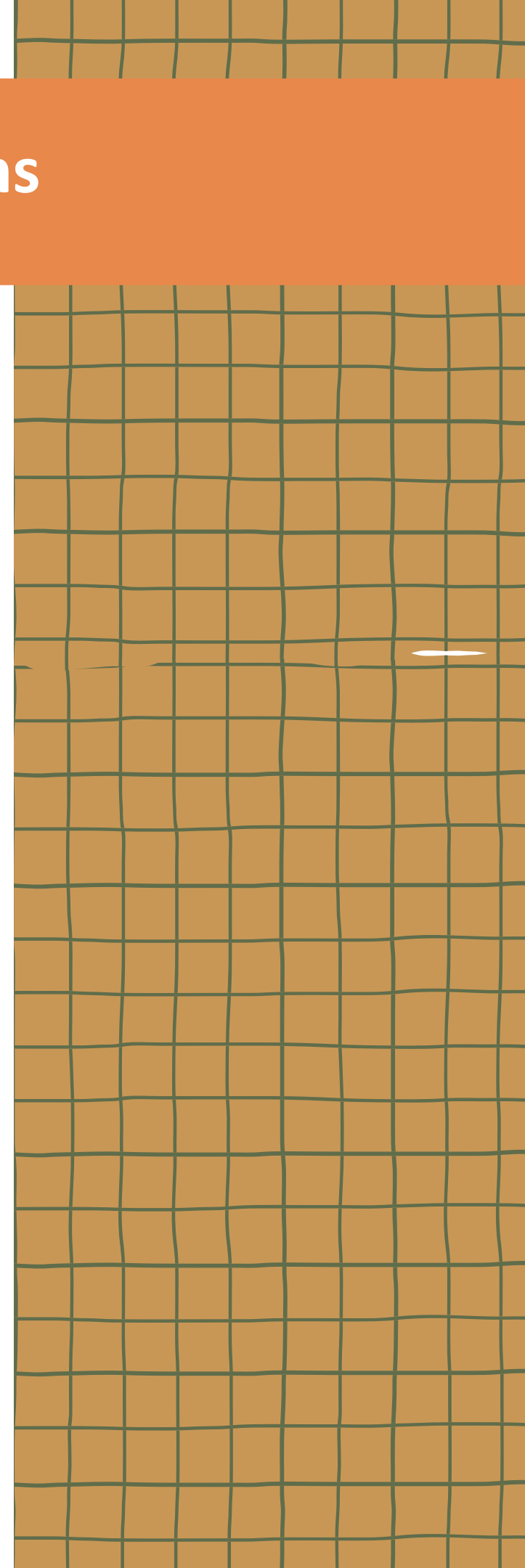
3. Draw the parabola:

- Sketch a smooth curve through the vertex and symmetric points.

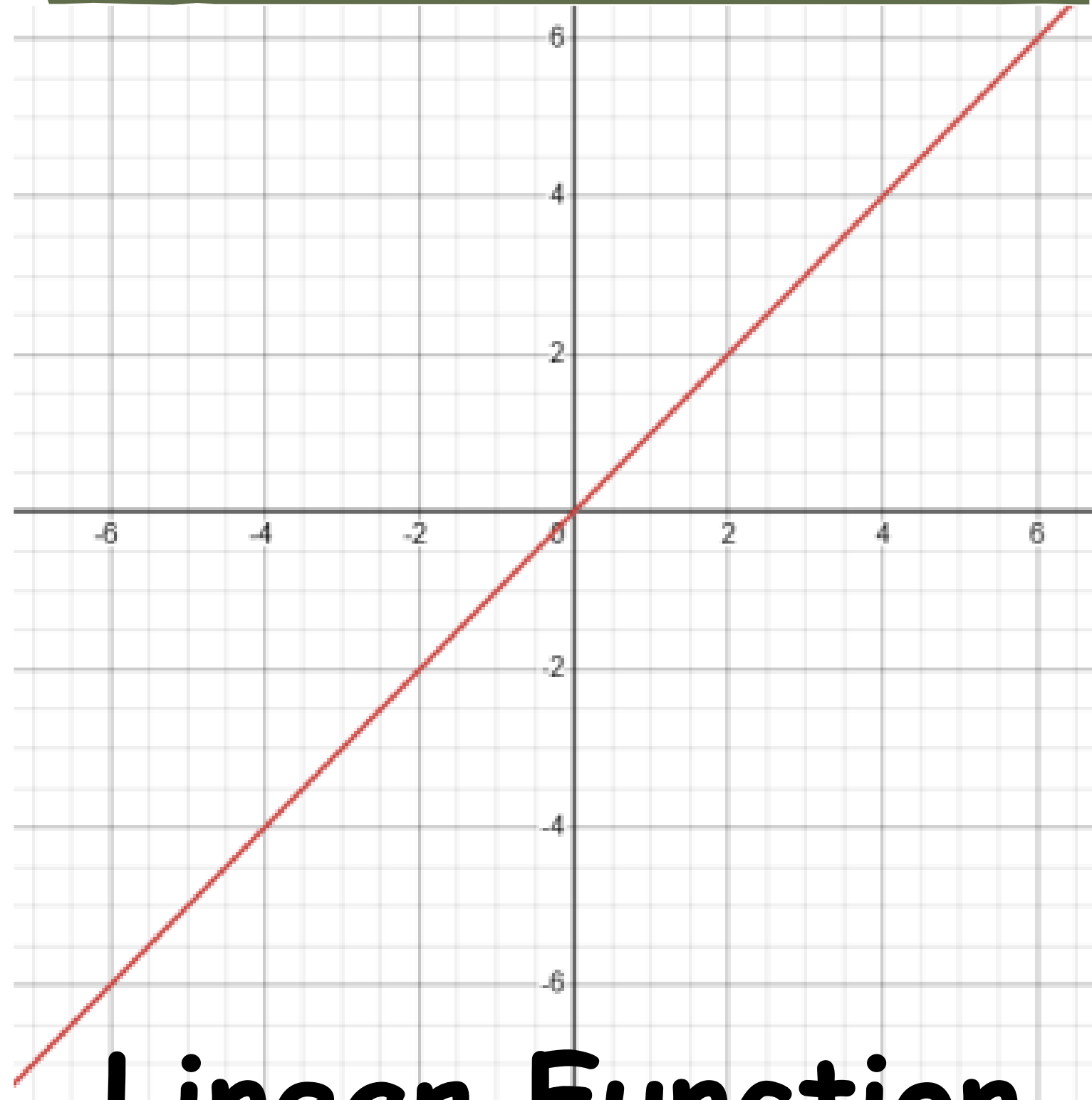
Here is the graph of the function $y = x^2 + 2$



Activity B.1: Classifying the Functions as Linear or Quadratic Through Graphs



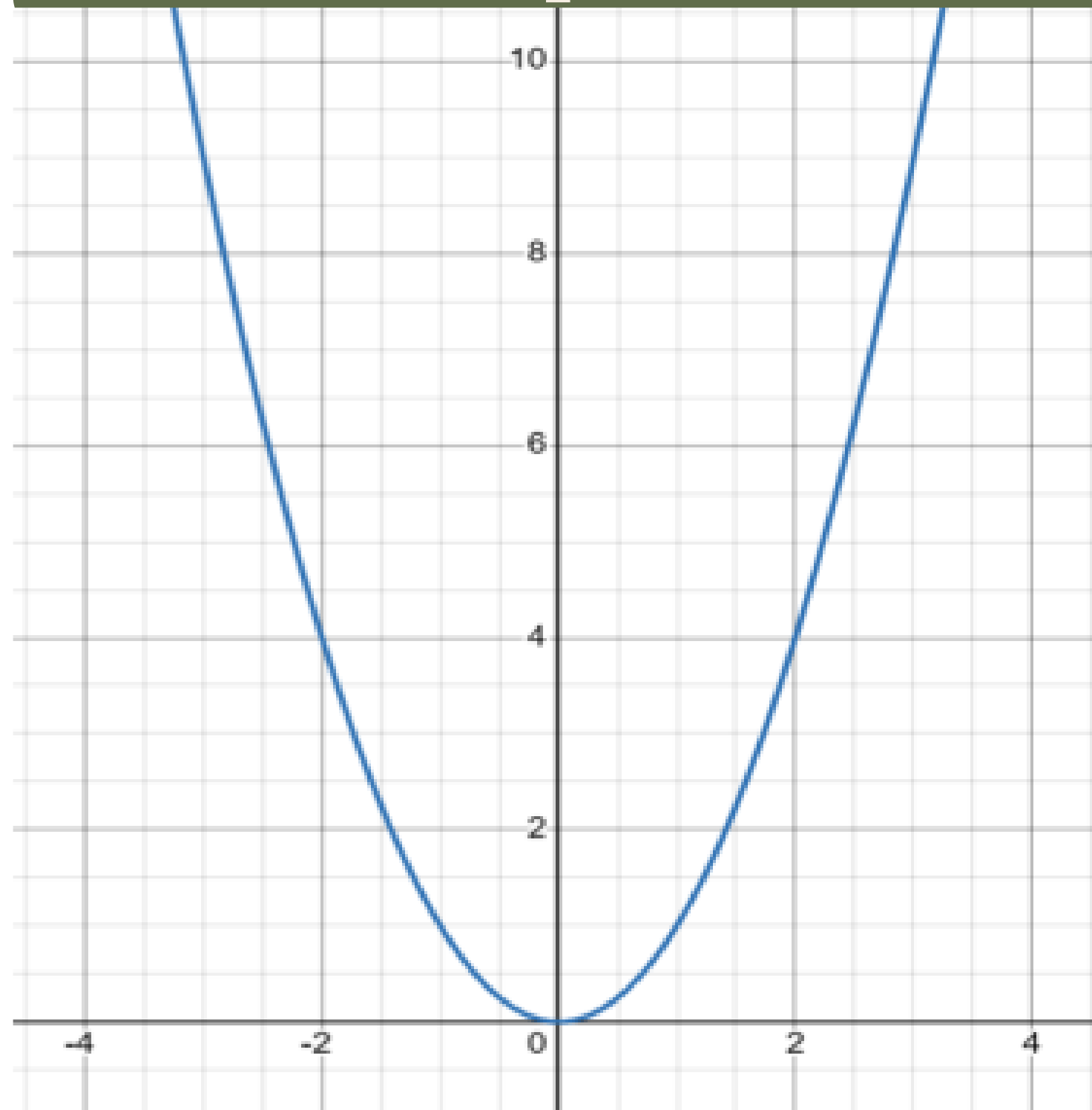
Graph 1



Linear Function



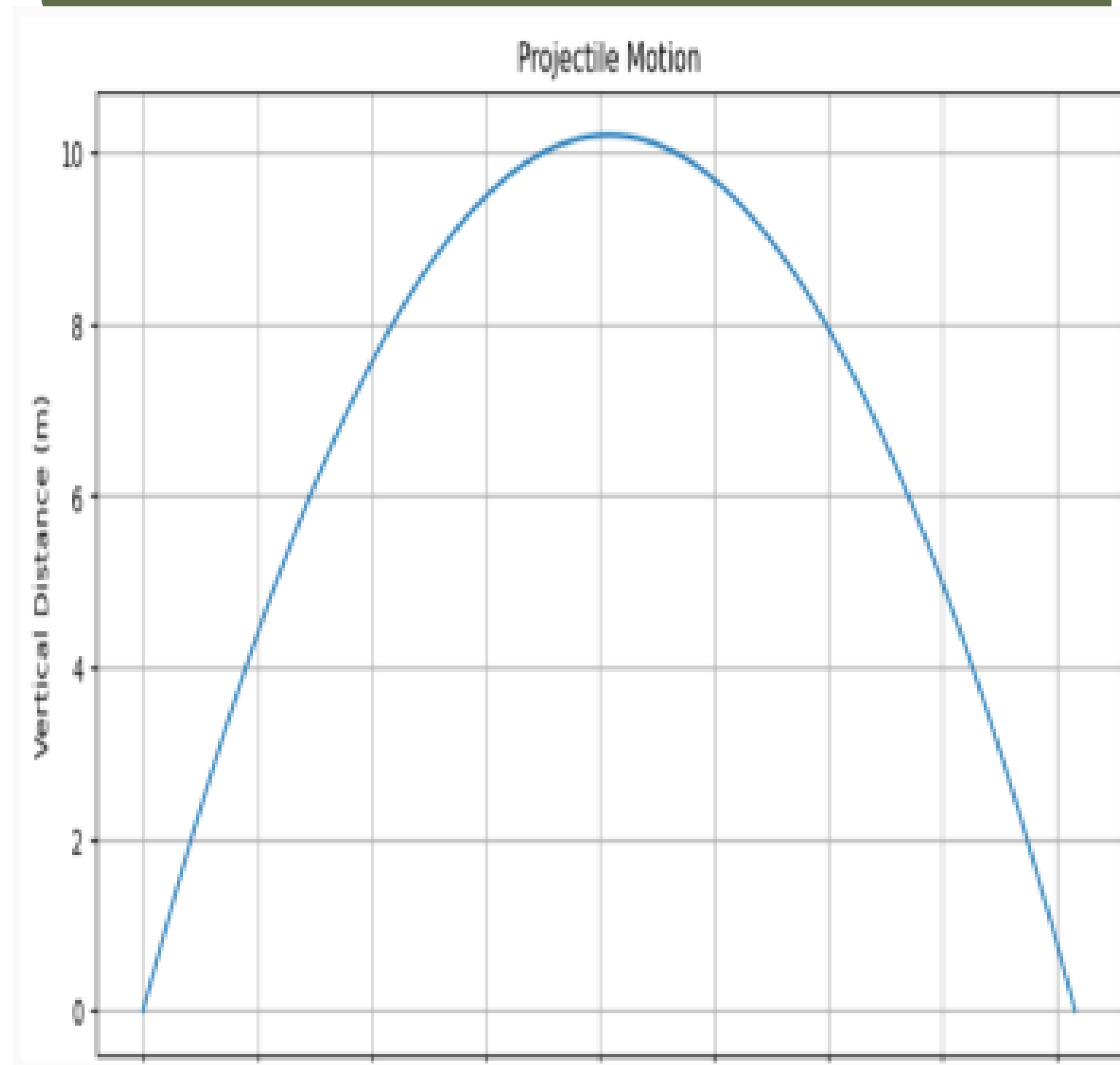
Graph 2



Quadratic Function



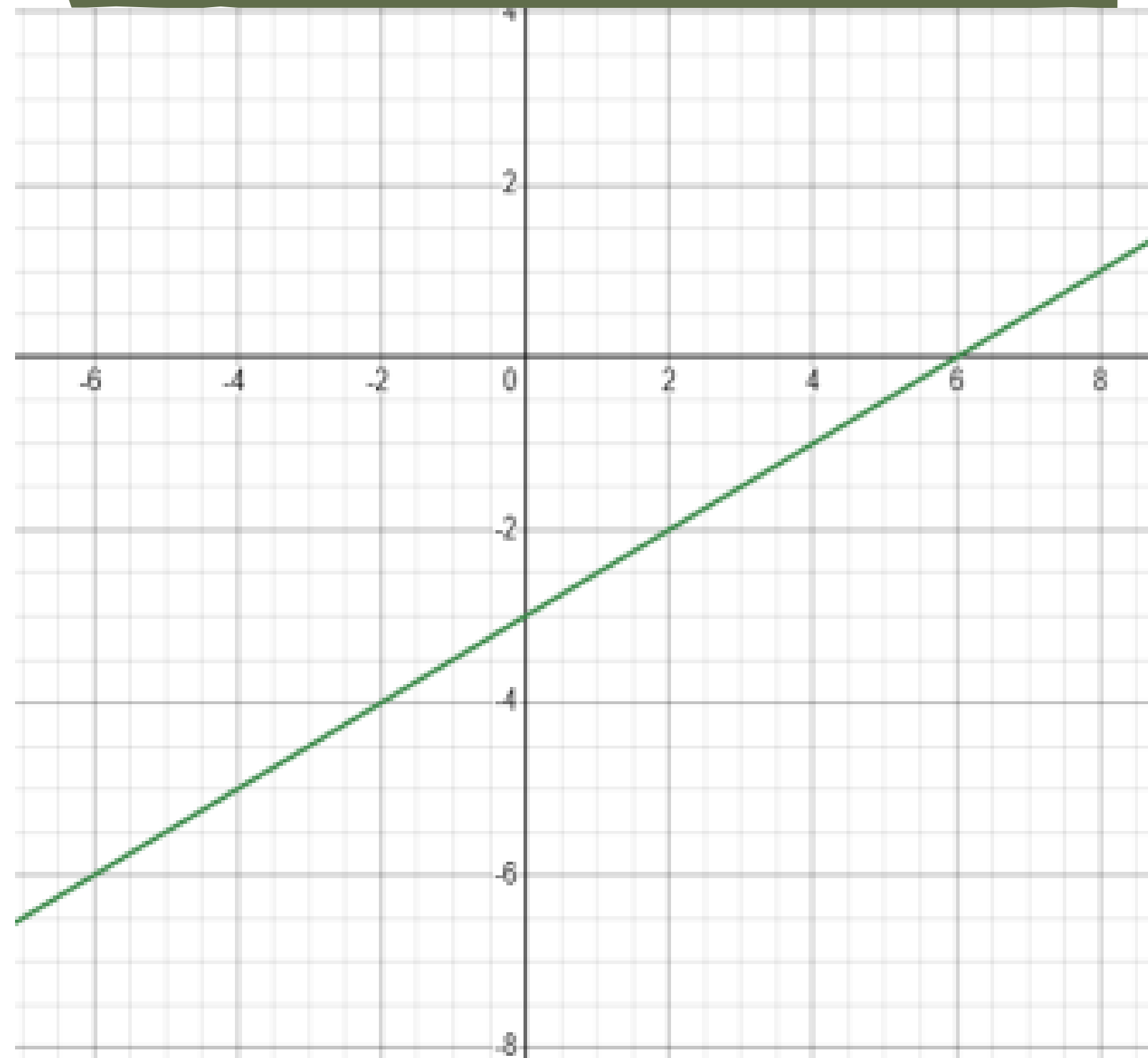
Graph 3



Quadratic Function



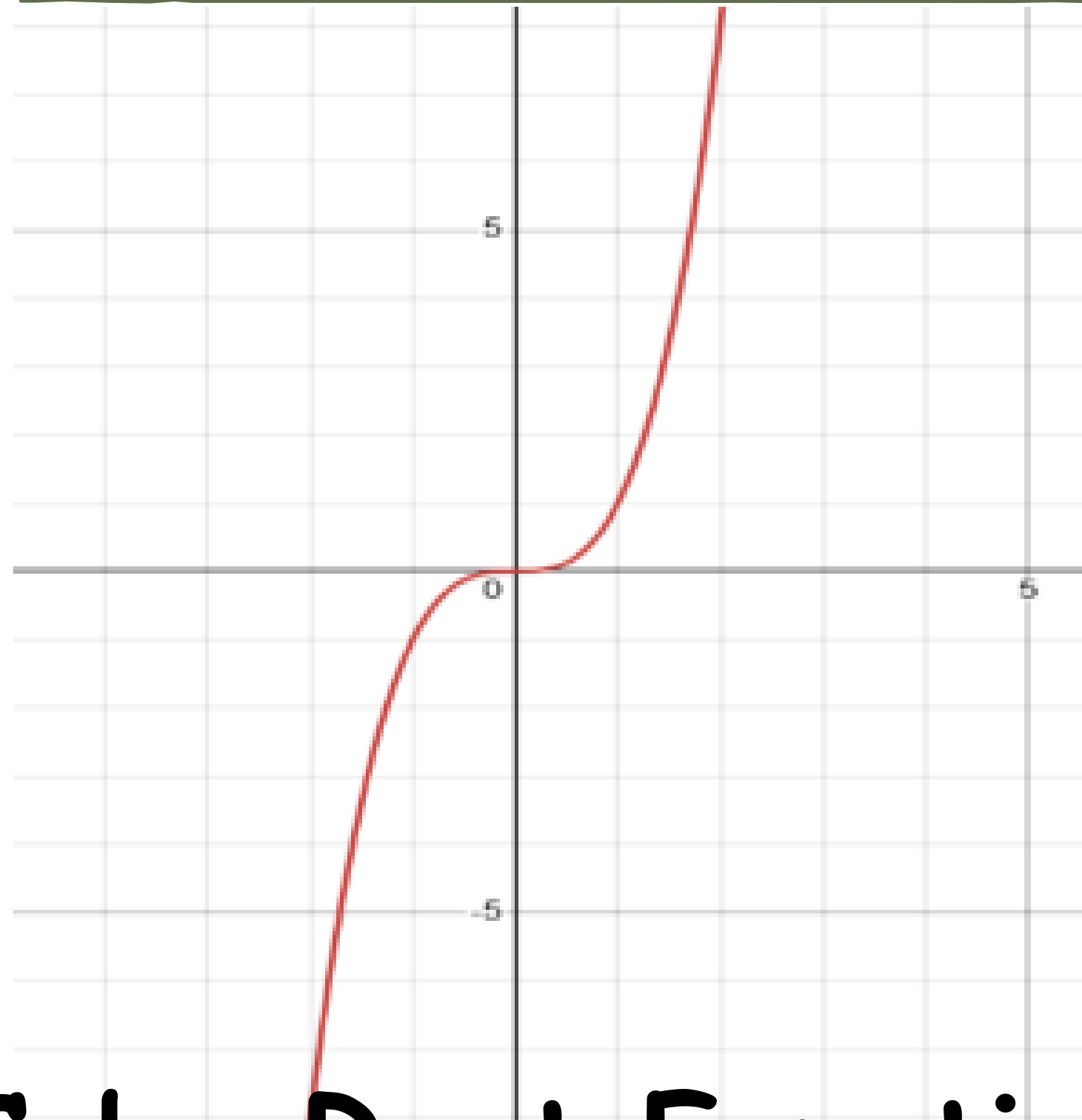
Graph 4



Linear Function



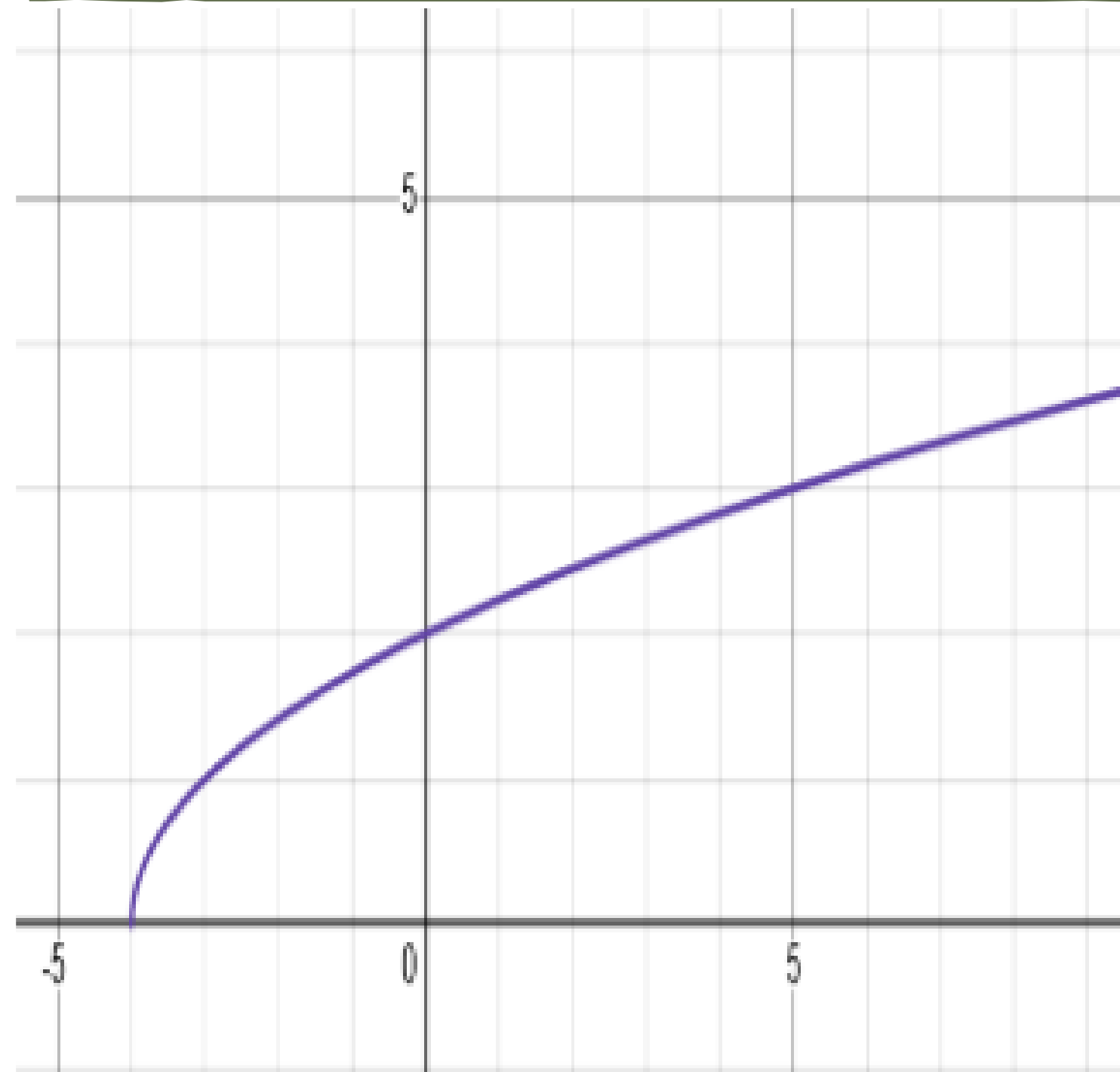
Graph 5



Cube Root Function



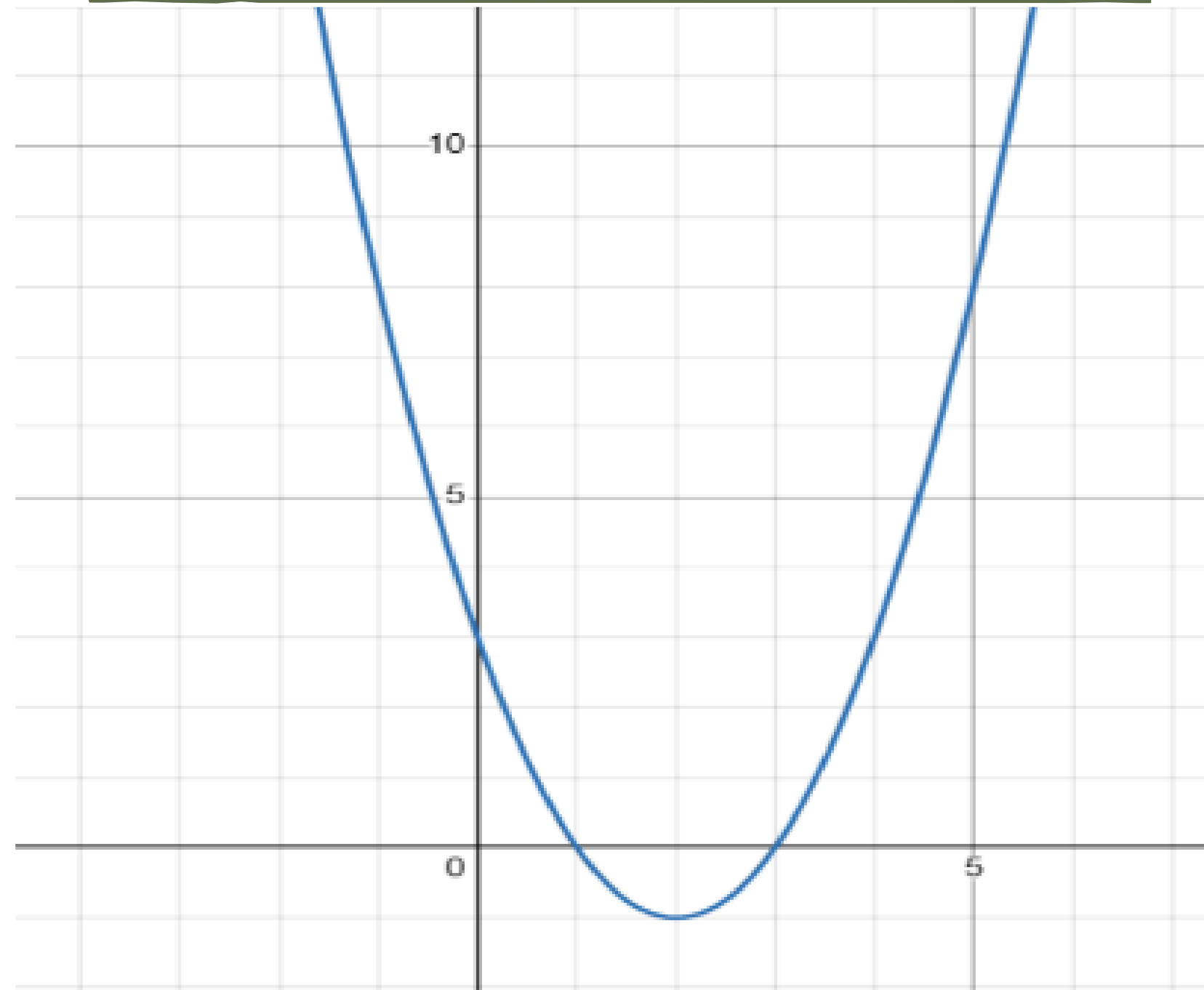
Graph 6



Square Root Function



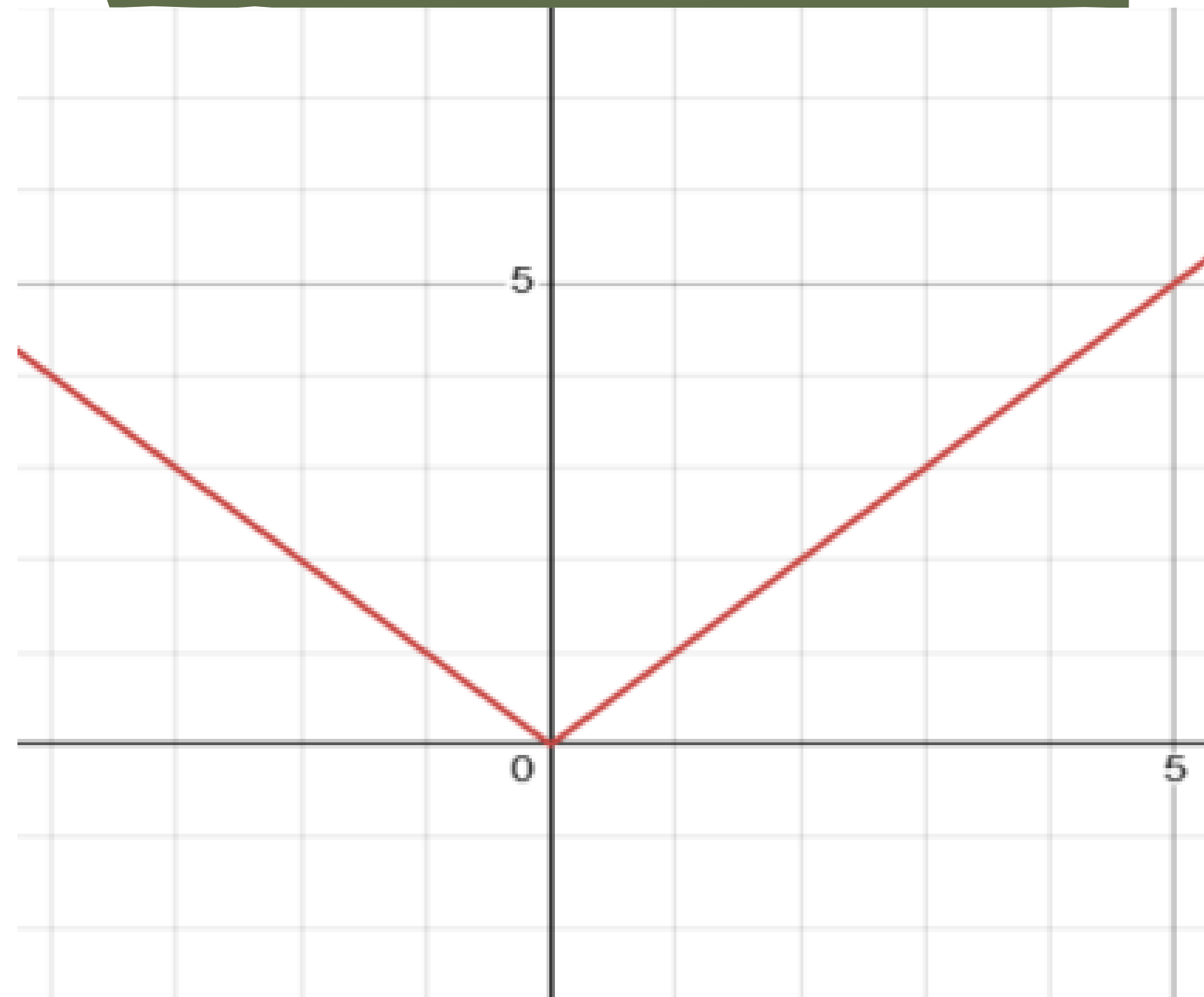
Graph 7



Quadratic Function



Graph 8



Absolute Value Function



Activity B.2

Lesson Discussion: Linear Functions and Quadratic Functions



Key Characteristics of Linear and Quadratic Functions

Linear Functions

1. Equation Form:

The general form of a linear function is $y = mx + b$, where:

*m is the **slope**.*

*b is the **y-intercept**.*

2. Graph Shape:

The graph of a linear function is a straight line.



Key Characteristics of Linear and Quadratic Functions

Linear Functions

3. Slope:

The slope (m) indicates the steepness and direction of the line.

Positive slope: Line rises from left to right.

Negative slope: Line falls from left to right.

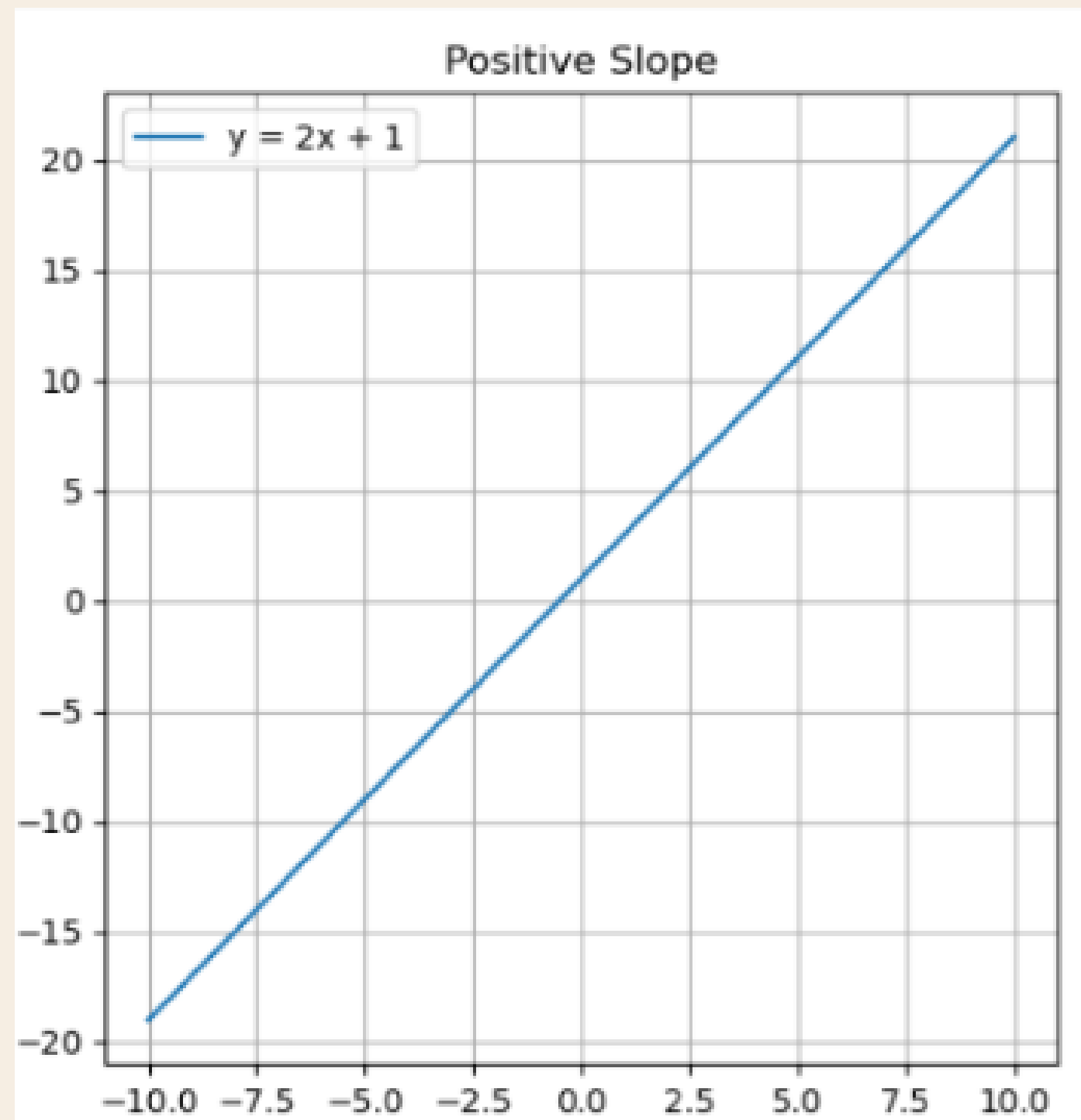
Zero slope: Line is horizontal.

Undefined slope: Line is vertical



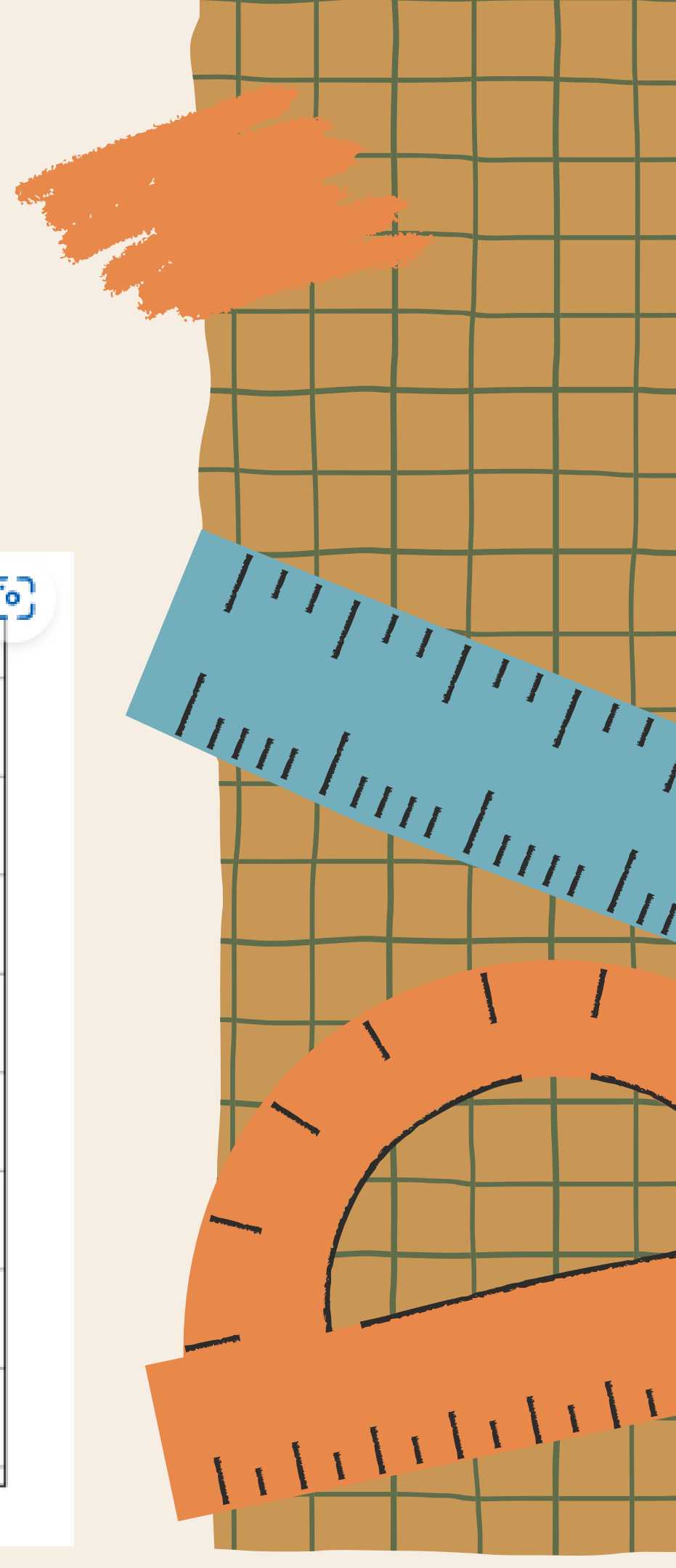
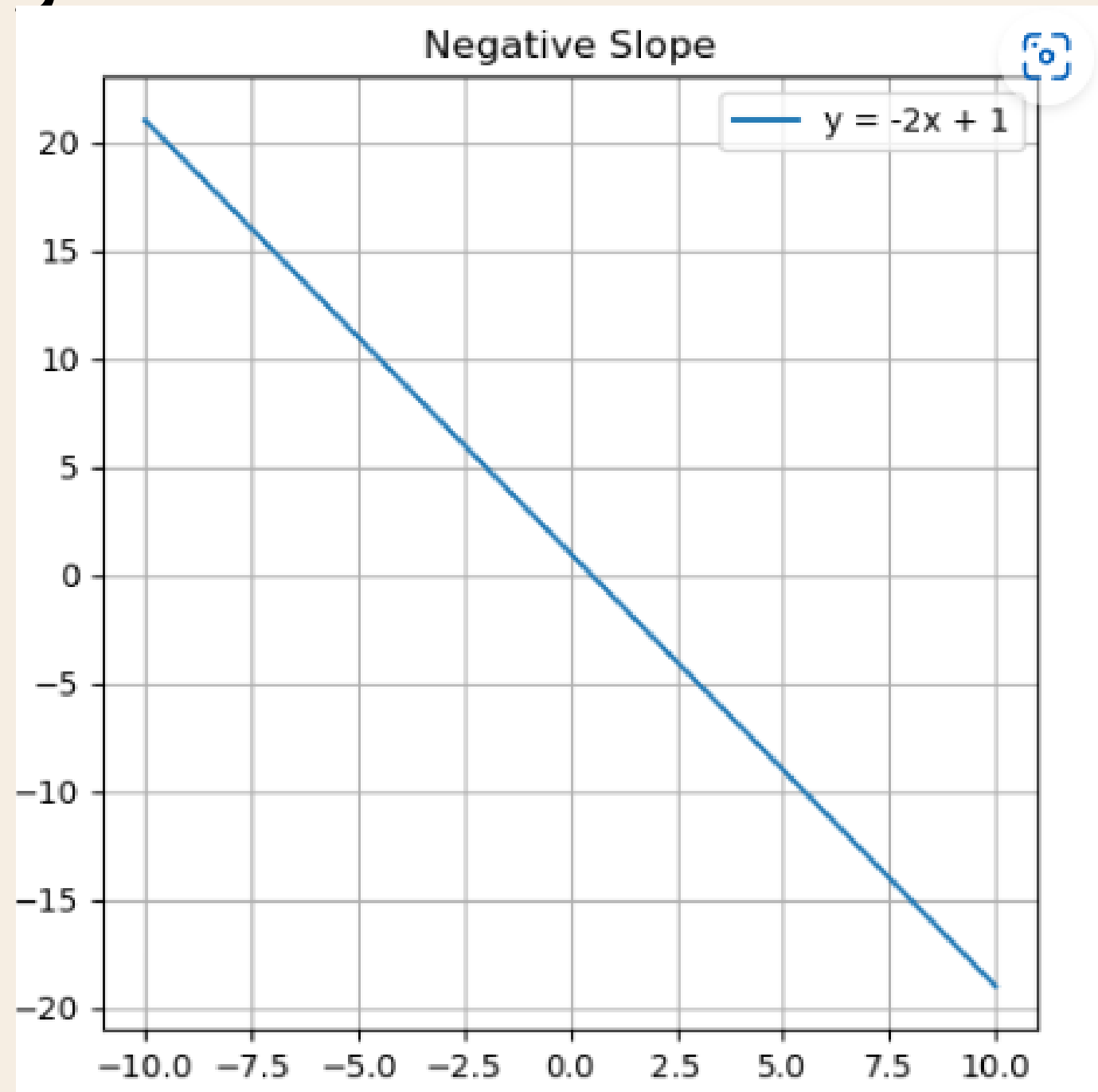
Example:
Linear function with positive slope: $y=2x+1$

The line rises from left to right.
Slope = 2, Y-intercept = 1.



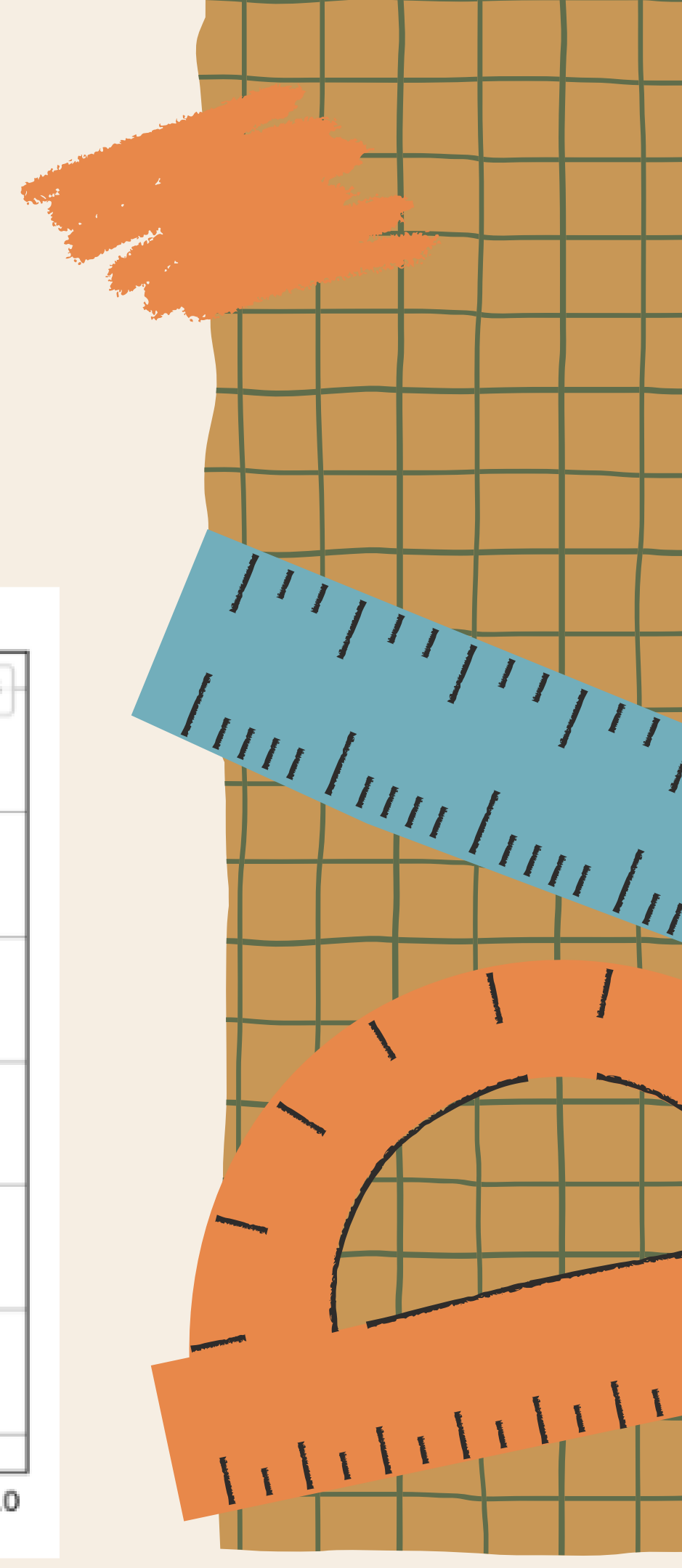
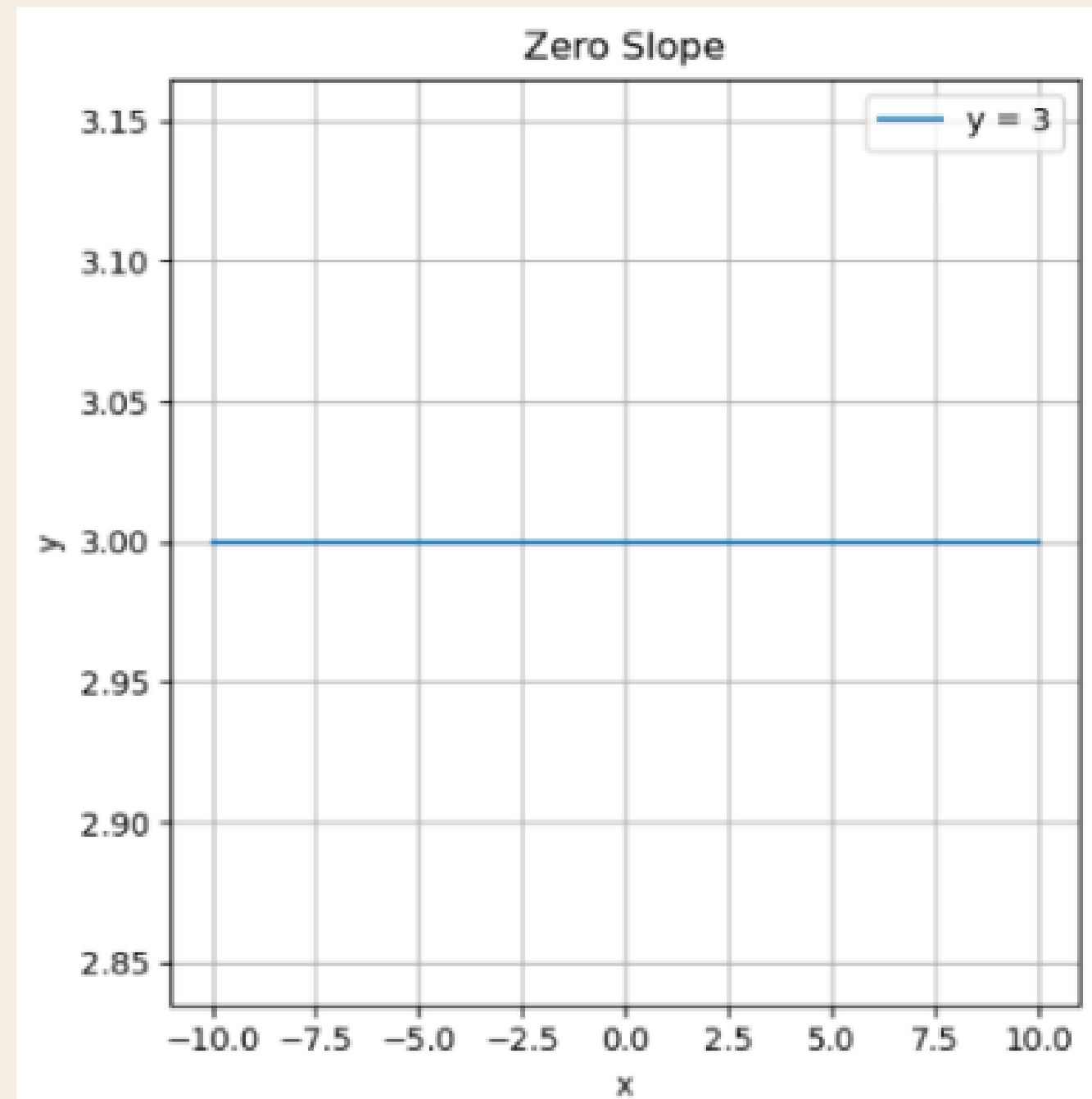
Example:
Linear function with negative slope:
 $y = -2x + 1$

The line falls from left to right.
Slope = -2, Y-intercept = 1.



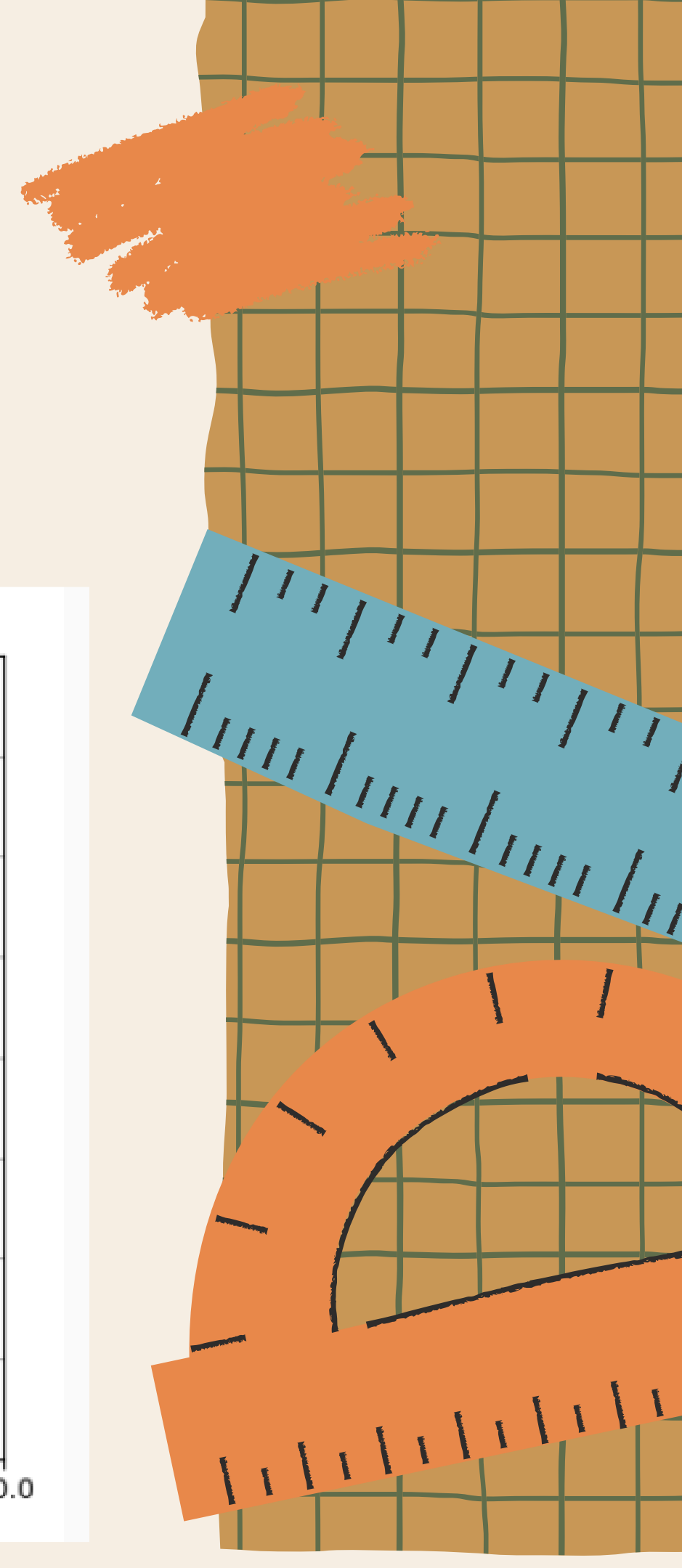
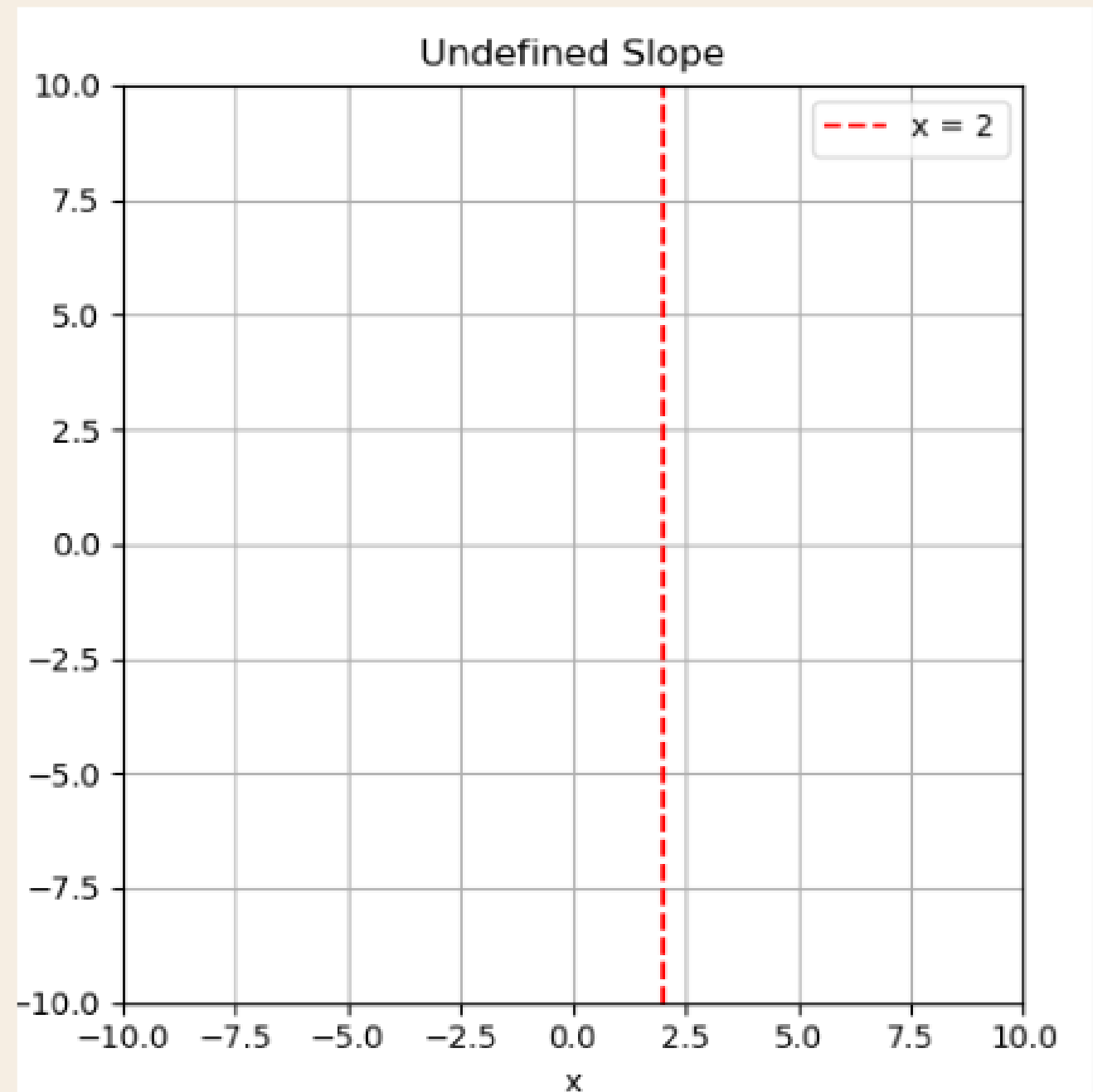
Example:
Linear function with zero slope: $y = 3$

The graph is a horizontal line.
The Slope = 0 and y-intercept = 3.



Example:
Linear function with undefined slope: $x = 2$

The graph is a vertical line.
Slope is undefined, passes through $x = 2$.



Key Characteristics of Linear and Quadratic Functions

Linear Functions

4. Y-Intercept:

The y -intercept (b) is the point where the line crosses the y -axis.

5. Rate of Change:

Linear functions have a constant rate of change.

The rate of change is the slope of the linear function.



Example 1: In the given function $y = 3x + 2$, find the slope and the y-intercept. Then sketch the graph.

Step 1: Identify the slope and y-intercept

Slope (m): The coefficient of (x) is 3.

Y-intercept (b): The constant term is 2, meaning the graph crosses the y-axis at $((0, 2))$.

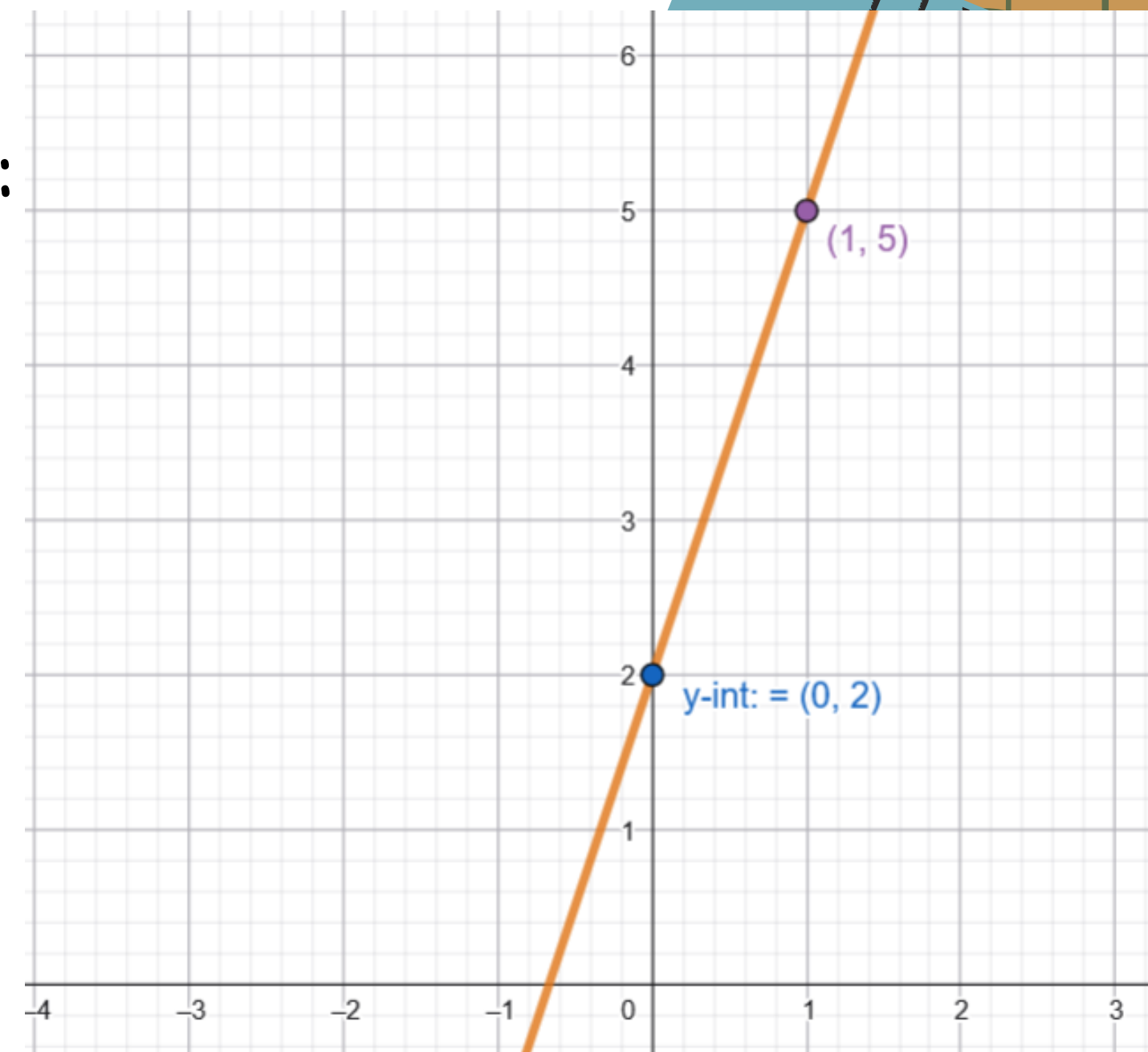
Step 2: Sketch the graph

Start at the **y-intercept** $(0, 2)$.

Use the **slope** to find another point:

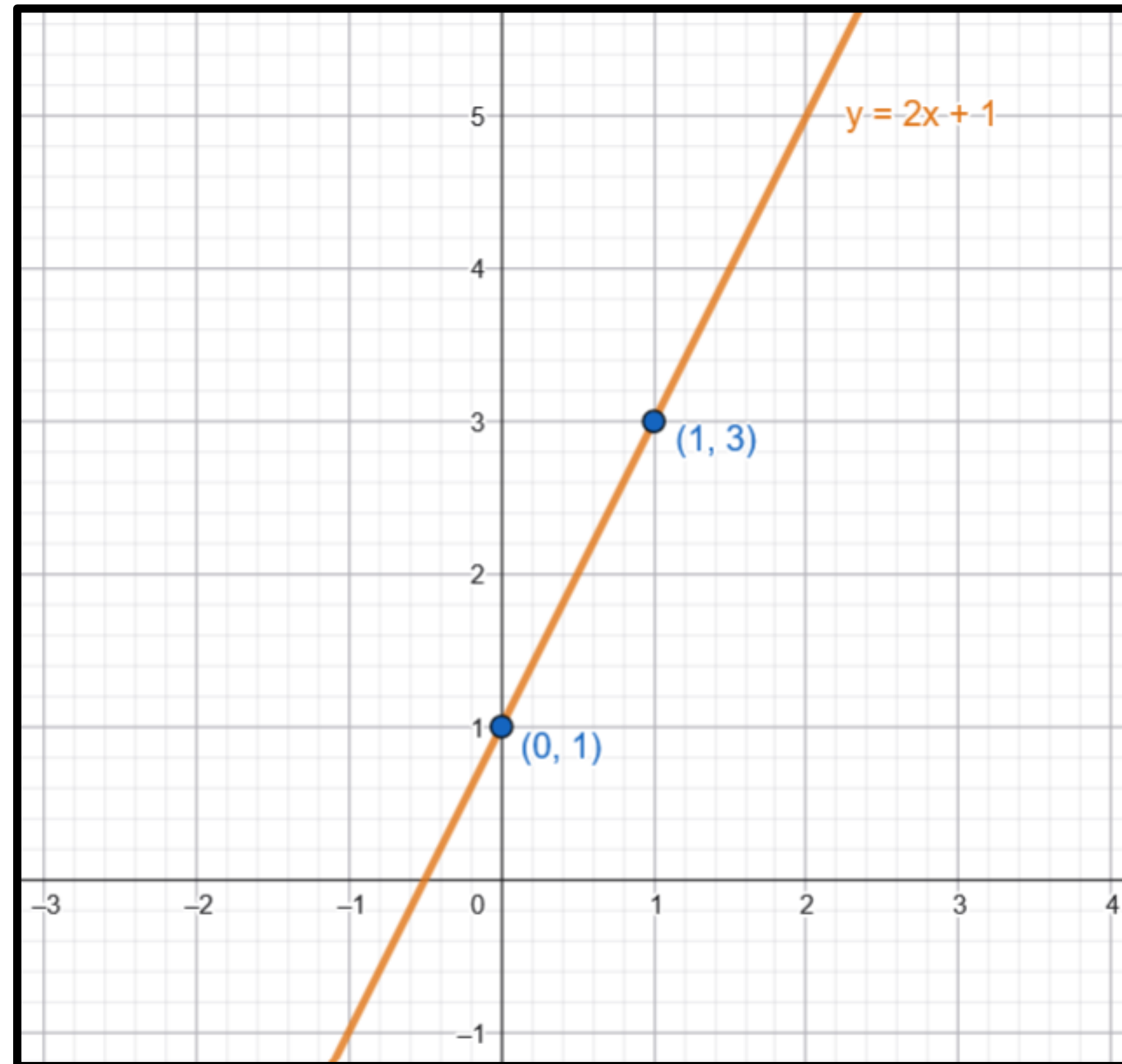
Since the slope is 3, go **up 3 units** and **right 1 unit** from the y-intercept to get the point $(1, 5)$.

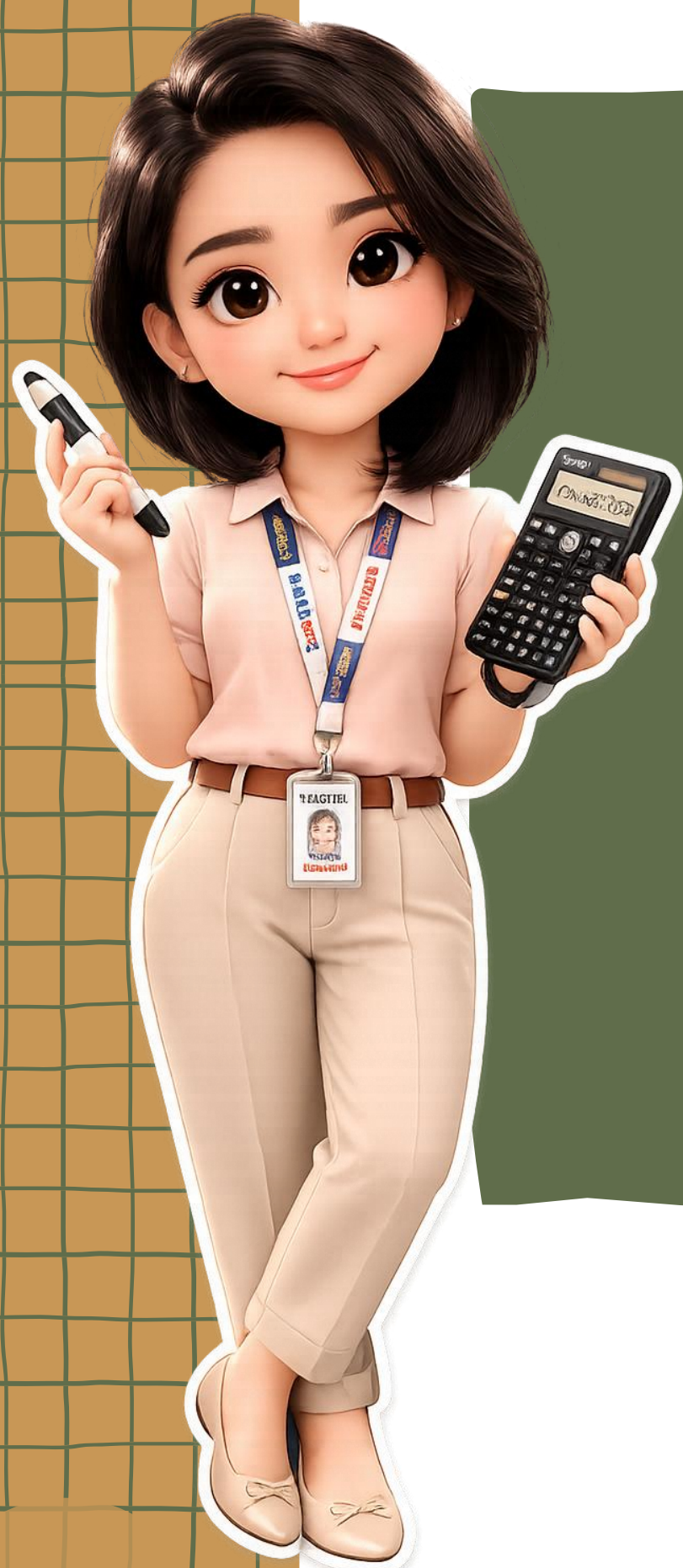
Draw a straight line through these points.



Example 2: Graph the function $y = 2x + 1$.

Answer:





Activity: YOUR TURN

Instructions:

Graph the following functions using the shortcut ways in graphing and answer the questions that follow.

a. $y = -2x + 2$

b. $y = 3x - 4$

c. $2x + y + 4 = 0$,

Guide Questions:

1. What do you notice on the graphs of the three functions?
2. How do you determine the points on the third function?

Discuss the Properties of Quadratic Functions

1. Equation Form:

The general form of a quadratic function is $y = ax^2 + bx + c$, where:

(a), (b), and (c) are constants and a is not equal to zero.

(a) determines the **direction** and **width** of the parabola.

2. Graph Shape:

The graph of a quadratic function is a **parabola**.

3. Vertex:

The **vertex** is the highest or lowest point on the graph, depending on the direction of the parabola.

If ($a > 0$), the parabola opens upwards, and the vertex is the minimum point.

If ($a < 0$), the parabola opens downwards, and the vertex is the maximum point.



Discuss the Properties of Quadratic Functions

4. Axis of Symmetry:

The axis of symmetry is a vertical line that passes through the vertex, given by $x = -b/2a$

5. Y-Intercept:

The y-intercept is the point where the parabola crosses the y-axis, found by evaluating (y) when (x = 0).

6. Direction and Width:

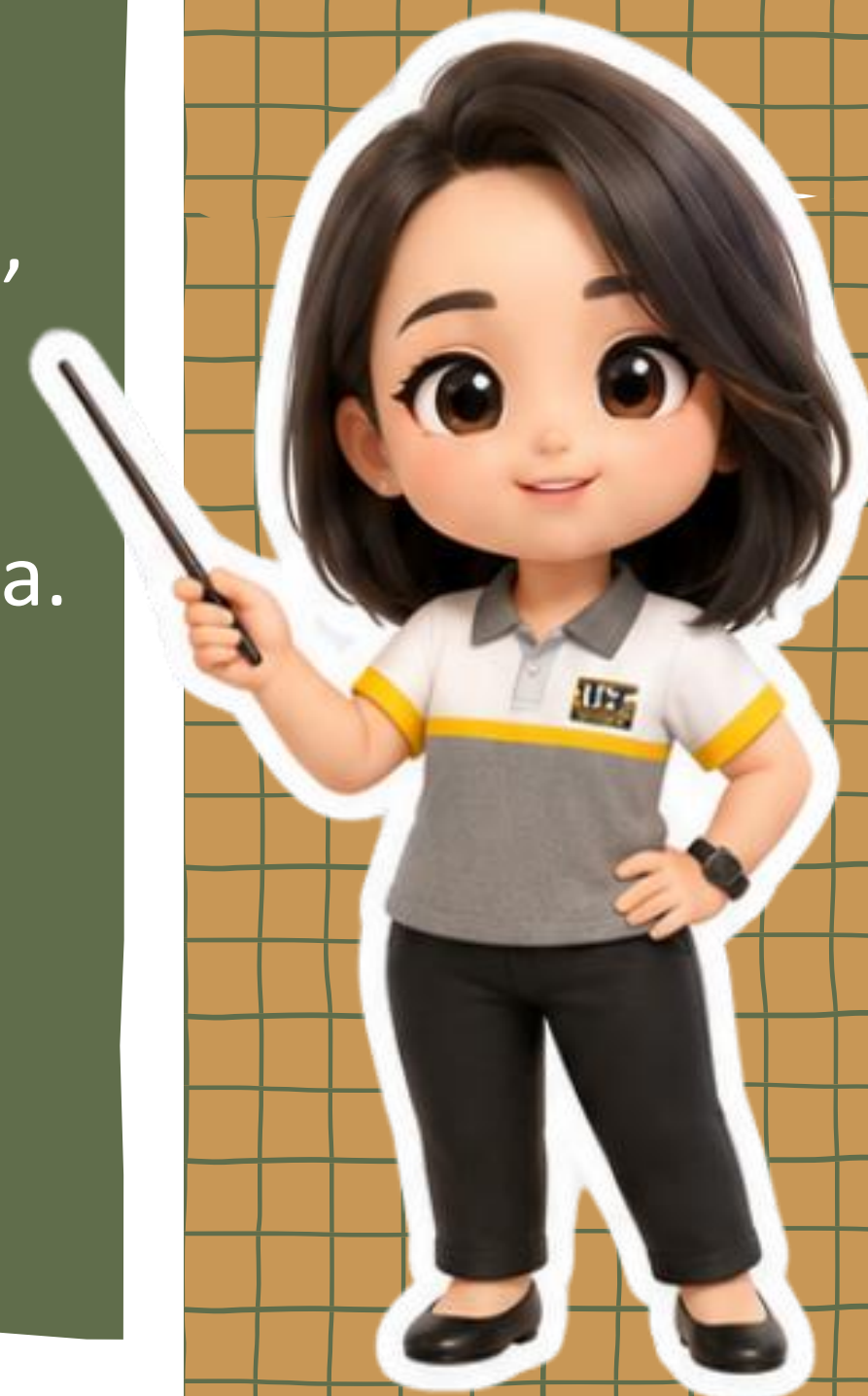
The coefficient (a) affects the direction and width of the parabola.

Larger absolute value of (a): **Narrower parabola.**

Smaller absolute value of (a): **Wider parabola.**

7. Roots/Zeros:

The roots or zeros of the quadratic function are the points where the graph intersects the x-axis, found by solving $ax^2 + bx + c = 0$.

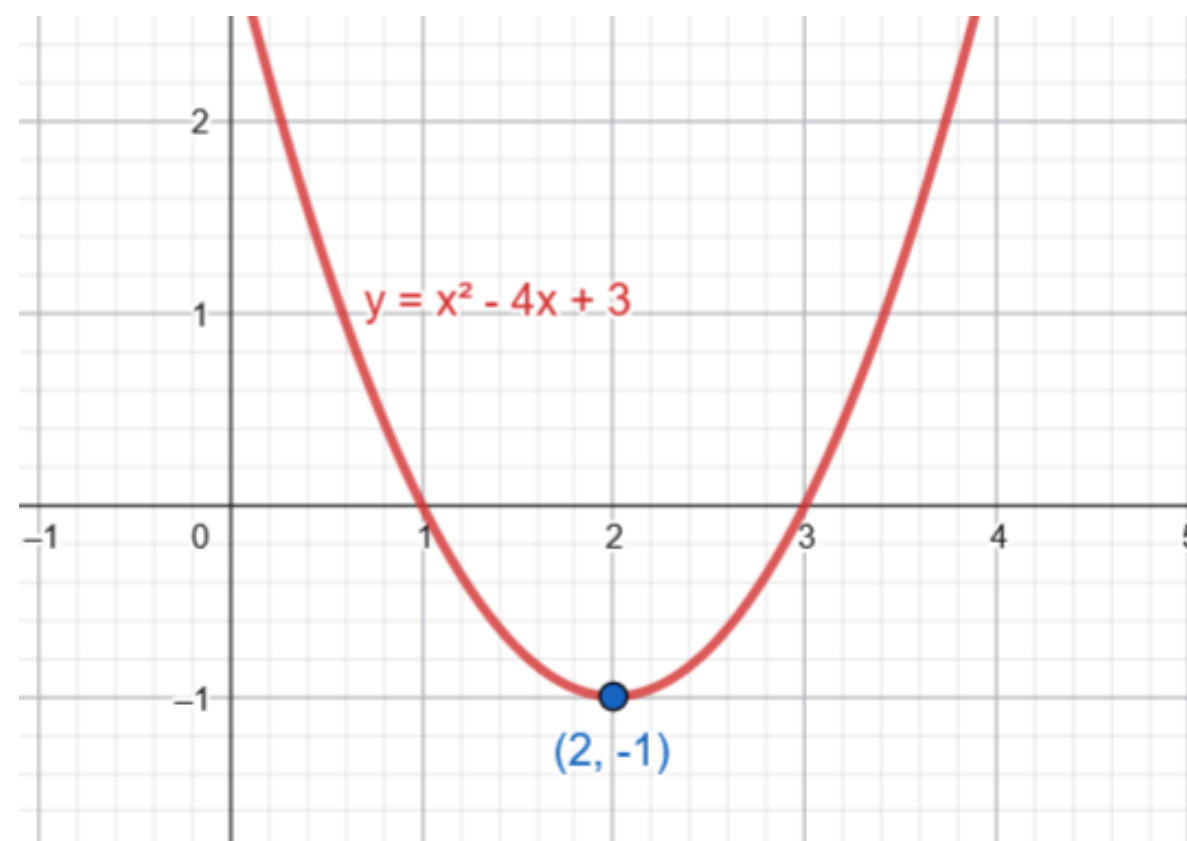


Example 3: Work in group to construct the graph of $y = x^2 - 4x + 3$ in large poster.

Instructions:

1. Divide the class into small groups of 3-5 learners.
2. In a Large Poster, graph the function $y = x^2 - 4x + 3$
3. You are free to use any techniques in graphing.
4. Each group presents their graph, explaining the process and features of the parabola.

Answer: Sample graph of $y = x^2 - 4x + 3$.



Vertex Form of a Quadratic Function

$$y = a(x-h)^2 + k$$

Where:

- (h, k) is the **vertex**
- a determines the **direction** and **width** of the parabola
- Shortcut Steps to Graph

Step 1: Identify the Vertex

- From the equation, extract h and k
- Example: $y = (x - 3)^2 + 2 \rightarrow$ Vertex is $(3, 2)$

Step 2: Plot the Vertex

- This is the turning point of the parabola.

Step 3: Determine the Direction

- If $a > 0$: Parabola opens **upward**
- If $a < 0$: Parabola opens **downward**

Step 4: Use “1-3-5” Rule for Standard Width

From the vertex, move: 1 unit left/right $\rightarrow$ 1 unit up/down

2 units left/right $\rightarrow$ 3 units up/down

3 units left/right $\rightarrow$ 5 units up/down

Multiply these values by ($|a|$) if ($a \neq 1$).

Step 5: Draw the Axis of Symmetry

A vertical line through ($x = h$)

Step 6: Sketch the Parabola

Plot symmetric points on both sides of the vertex

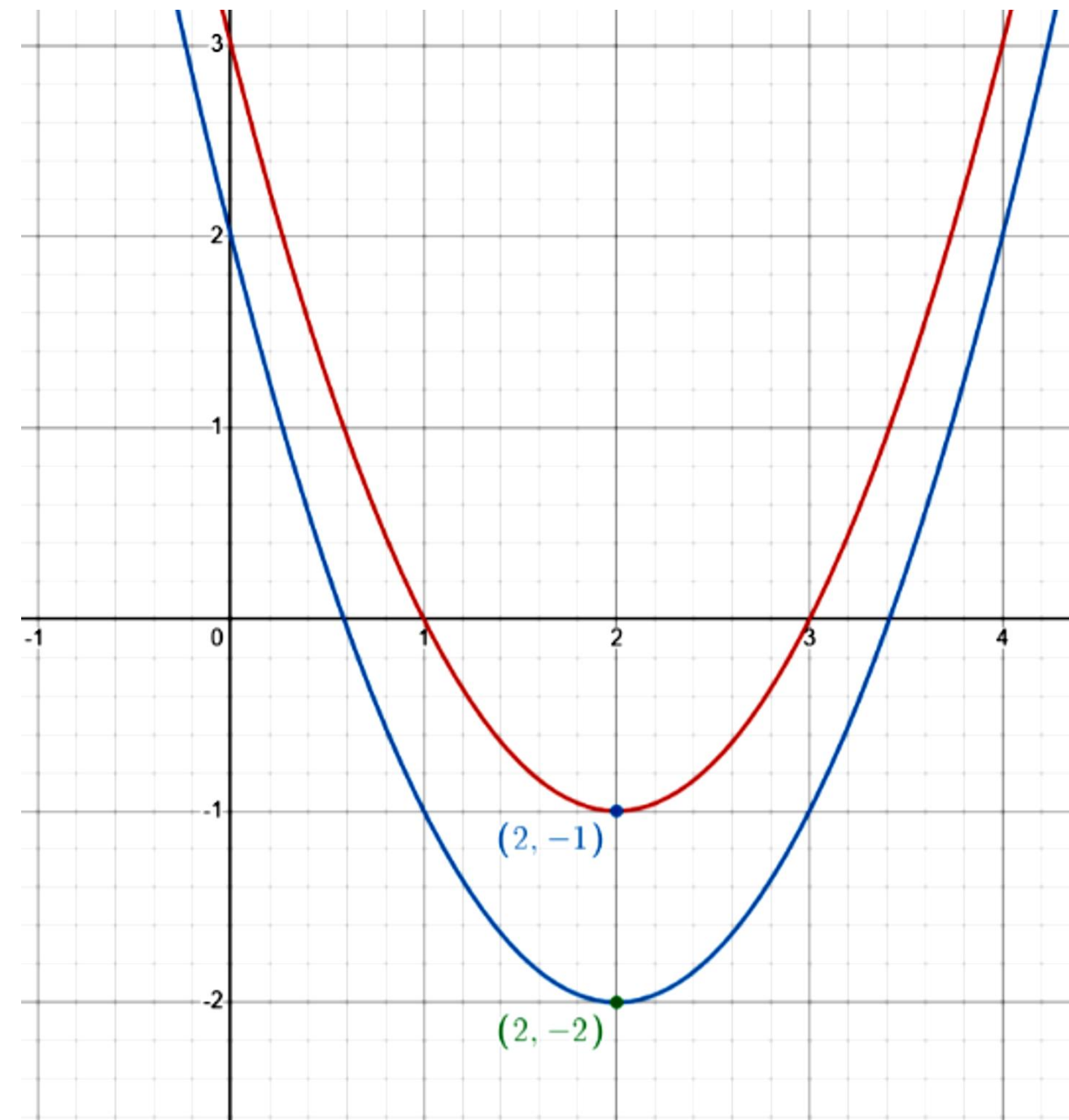
Connect the points with a smooth curve

Note: if quadratic equation in standard form $y=ax^2+bx+c$, convert into vertex form $y=a(x-h)^2+k$.

Show the steps of converting standard form into vertex form. Watch the steps in the YouTube video: Introduction to vertex form of a quadratic https://www.youtube.com/watch?v=_QqhuLixNEk

Example 4: Graph the function $y = x^2 - 4x + 2$. Then compare the graph of $y = x^2 - 4x + 3$.

Answer:



The comparison of the graphs of
 $y = x^2 - 4x + 2$ and $y = x^2 - 4x + 3$:

A. Vertex:

- Both functions have the same vertex x - coordinate, which is $x = 2$.
- The y-coordinates of the vertices are different:
For $y = x^2 - 4x + 2$, the vertex is at $(2, -2)$
For $y = x^2 - 4x + 3$, the vertex is at $(2, -1)$

B. Shape and Direction:

- Both graphs are parabolas that open upwards because the coefficient a is positive in both cases.

C. Y-intercept

The y-intercepts are different:

- For $y = x^2 - 4x + 2$, the y-intercept is $(0, 2)$
- For $y = x^2 - 4x + 3$, the y-intercept is $(0, 3)$



Lesson Discussion:

Absolute Value Function, Square
Root Function and Cube Root
Function

Key Characteristics

Absolute Value Function

General form: $f(x) = a|x - h| + k$, where (h, k) is the vertex of the V-shape.

From the equation: The variable x is enclosed inside absolute value bars $| |$.

From the graph: The graph forms a sharp V-shape (or an upside-down V if a is negative), with a single pointed vertex where the two straight line segments meet.

Domain and Range: The domain is all real numbers. The range is $y \geq k$ if $a > 0$, or $y \leq k$ if $a < 0$.



Key Characteristics

Absolute Value Function

Key Characteristics

- ❑ Domain: All real numbers
- ❑ Range: $y \geq k$ if $a > 0$, or $y \leq k$ if $a < 0$.
- ❑ Graph shape: a V-shaped graph with a sharp corner (vertex) at (h, k) , symmetric about the vertical line $x = h$.



Absolute Value Function

Steps in Graphing

Step 1: Identify the Vertex

- From the equation, extract (h) and (k)
- Example: $f(x) = |x - 2| + 1 \rightarrow$ Vertex is ($2, 1$)

Step 2: Plot the Vertex

Step 3: Determine the Direction, Domain & the Range

- If ($a > 0$): Parabola opens upward
- If ($a < 0$): Parabola opens downward

Step 4: Construct Table of Values

- Choose two x -values on each side of h , complete the table of values.

Step 5: Plot the points

Connect the points with two straight rays forming a V-shape, opening upward or downward based on the sign of a .

Absolute Value Function

Sketch the Graph: $f(x) = |x - 2| + 1$

Step 1: Identify the Vertex

- Vertex: (2,1)

Step 2: Plot the Vertex

Step 3: Determine the Direction

- If ($a > 0$): Parabola opens upward

Axis of Symmetry: $x = 2$

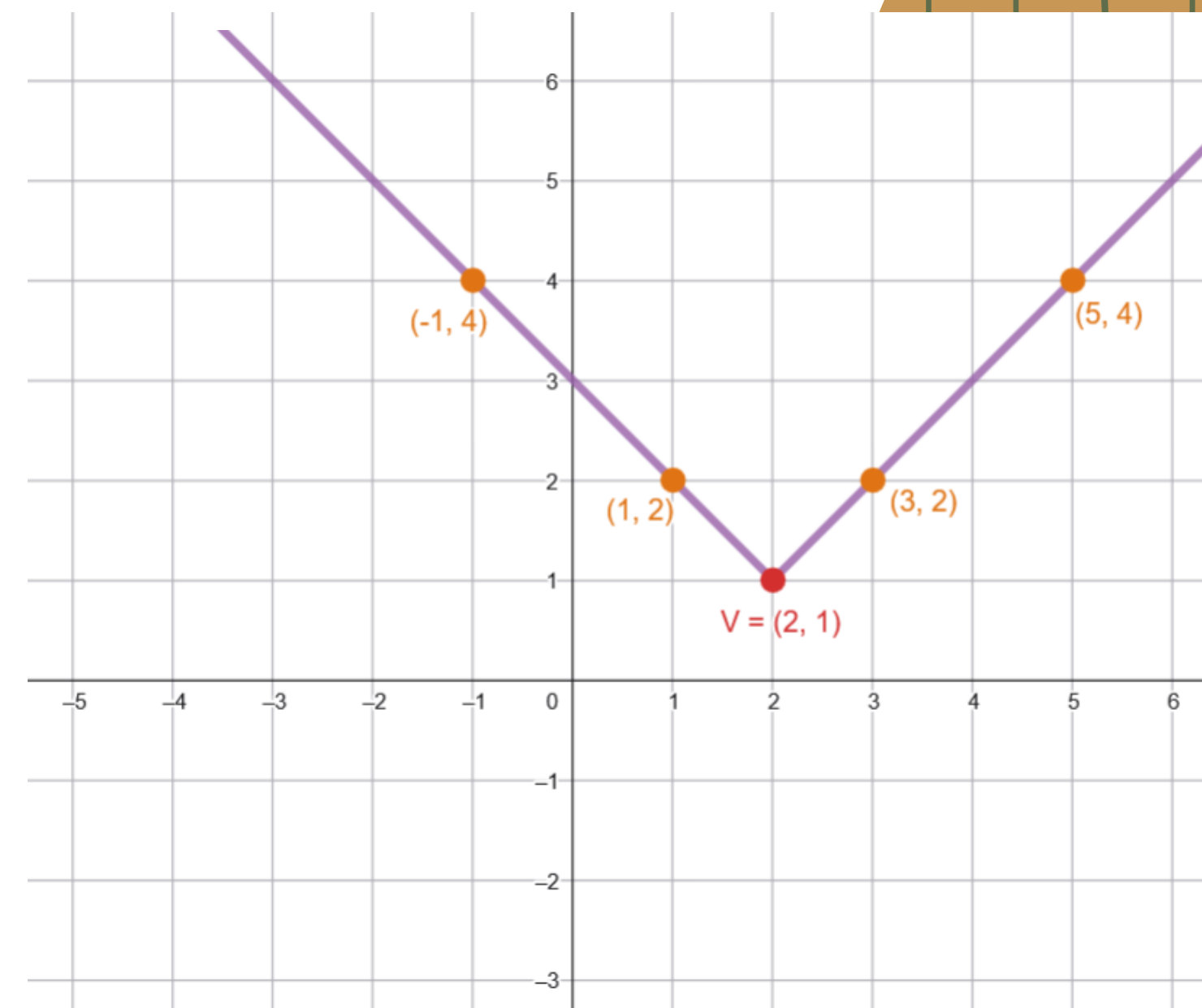
Domain: All Real Numbers

Range: $y \geq 1$

Step 4: Construct Table of Values

x	-1	1	2	3	5
y	4	2	1	2	4

Step 5: Plot the points



Absolute Value Function

Sketch the Graph: $f(x) = -|x + 2| + 4$

Step 1: Identify the Vertex

- Vertex: $(-2, 4)$

Step 2: Plot the Vertex

Step 3: Determine the Direction

- If $(a < 0)$: Parabola opens downward

Axis of Symmetry: $x = -2$

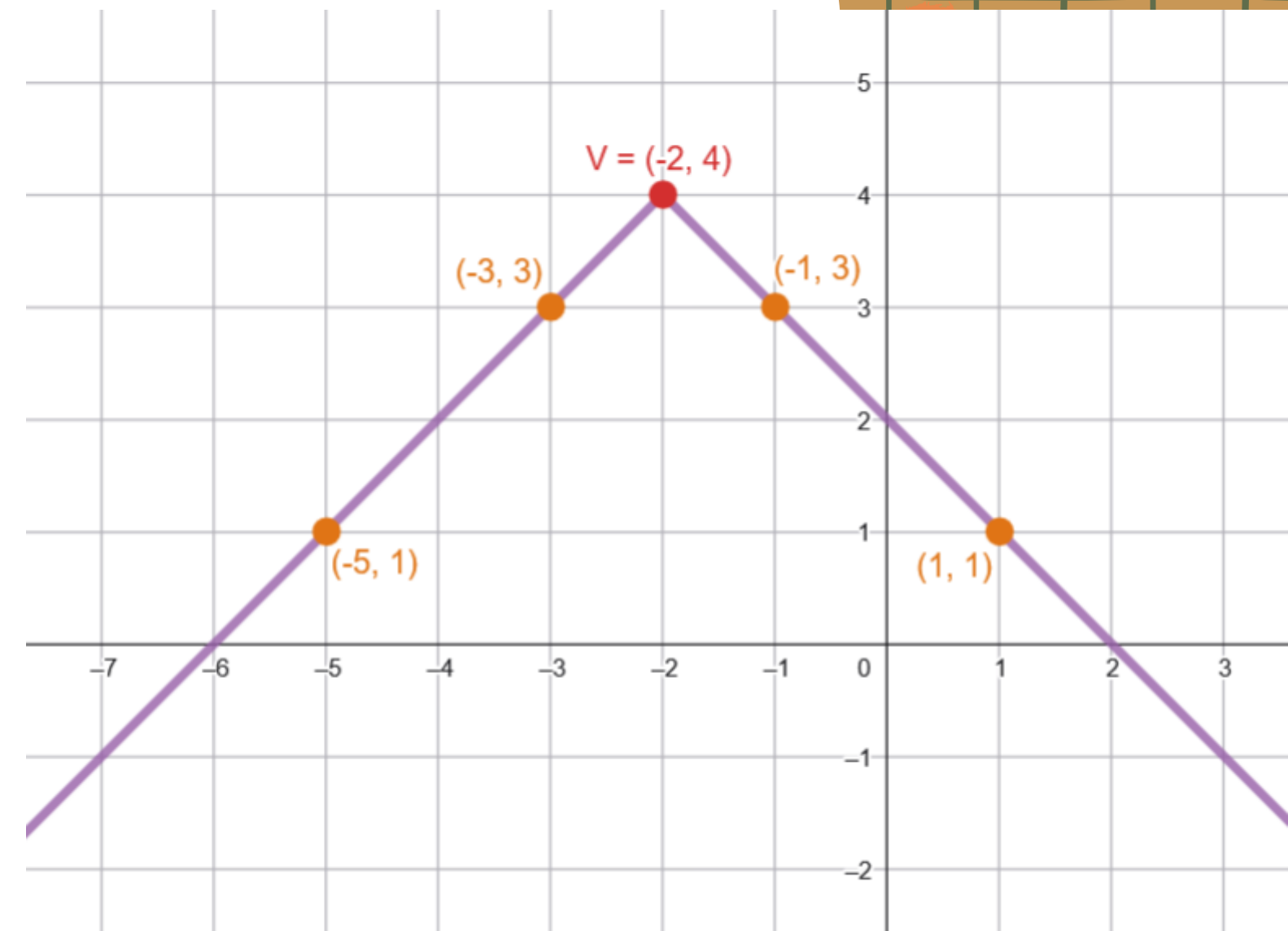
Domain: All Real Numbers

Range: $y \leq 4$

Step 4: Construct Table of Values

x	-5	-3	-2	-1	1
y	1	3	4	3	1

Step 5: Plot the points



Absolute Value Function

Sketch the Graph: $f(x) = 2|x - 1| + 1$

Step 1: Identify the Vertex

- Vertex: (1,1)

Step 2: Plot the Vertex

Step 3: Determine the Direction

- If ($a > 0$): Parabola opens upward

Axis of symmetry: $x = 1$

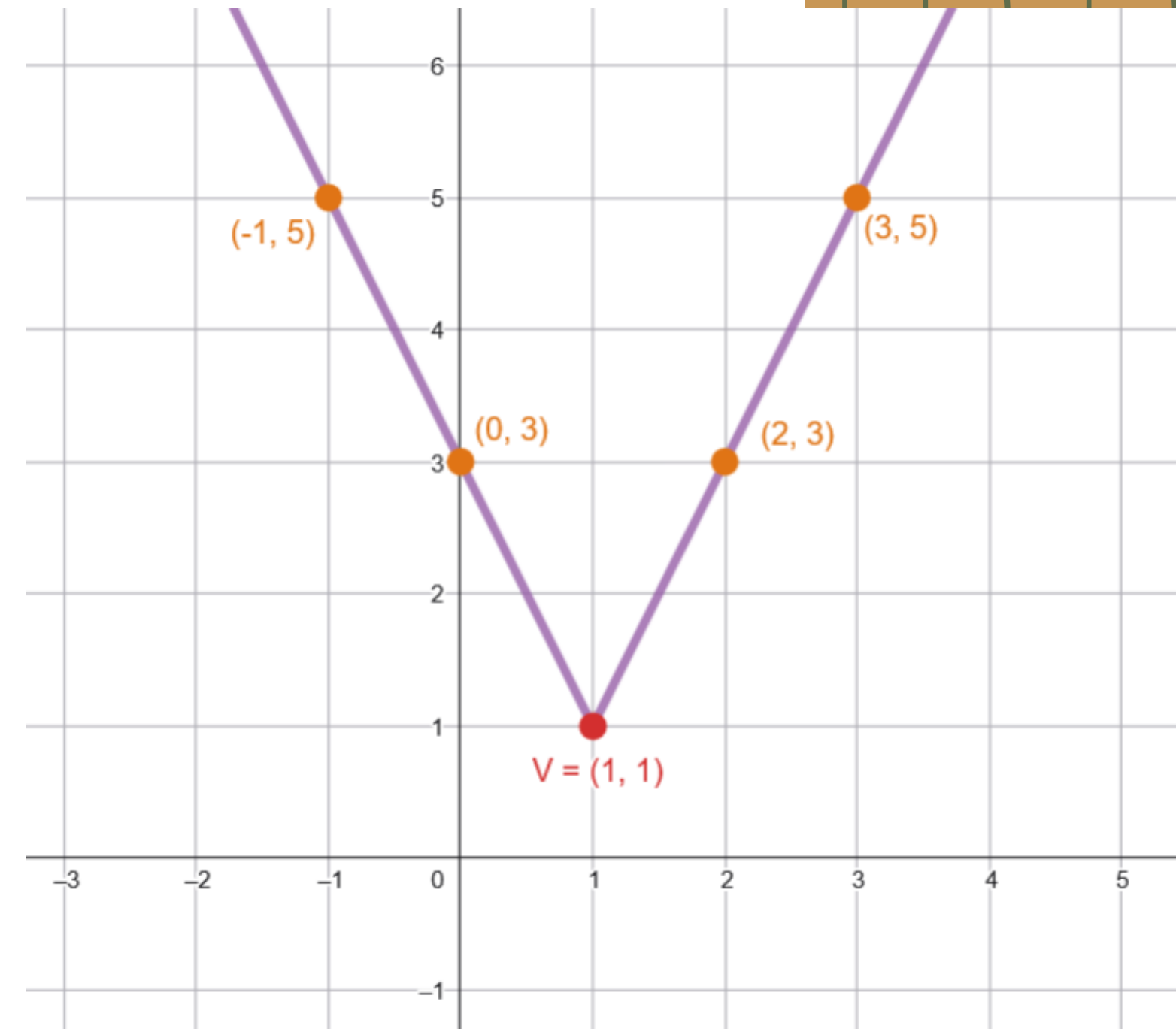
Domain: All Real Numbers

Range: $y \geq 1$

Step 4: Construct Table of Values

x	-1	0	1	2	3
y	5	3	1	3	5

Step 5: Plot the points



Key Characteristics

Square Root Function

General form: $f(x) = a\sqrt{x - h} + k$.

From the equation: The variable x appears underneath a radical sign ($\sqrt{}$) with an implied index of 2 (a square root).

From the graph: The graph looks like half of a sideways parabola — it starts at a single endpoint and curves gently upward or downward, never doubling back.

Domain and Range: The domain is restricted to $x \geq h$, since the radicand (the expression inside the square root) cannot be negative. The range is likewise restricted, typically $y \geq k$ or $y \leq k$.



Key Characteristics

Square Root Function

Key Characteristics

- ❑ Domain: $x \geq h$ (the value that keeps the radicand non-negative).
- ❑ Range: $y \geq k$ if $a > 0$, or $y \leq k$ if $a < 0$.
- ❑ Graph shape: half of a sideways parabola, starting at the point (h, k) and curving gradually outward.



Square Root Function

Steps in Graphing

Step 1: Determine the starting point (h, k),
which marks the beginning of the domain

- From the equation, extract (h) and (k)
- Example: $f(x) = \sqrt{x - 1} + 2 \rightarrow$ Starting point ((1, 2))

Step 2: Determine the Domain and the Range

Step 3: Construct Table of Values

Choose two or three x-values greater than h that make the radicand a perfect square for easy computation.

Step 4: Plot the points

Plot the resulting points and connect them with a smooth curve moving away from the starting point.

Square Root Function

Sketch the Graph: $f(x) = \sqrt{x - 1} + 2$

Step 1: Determine the starting point (h, k),
Starting point (1, 2)

Step 2: Determine the Domain and the Range

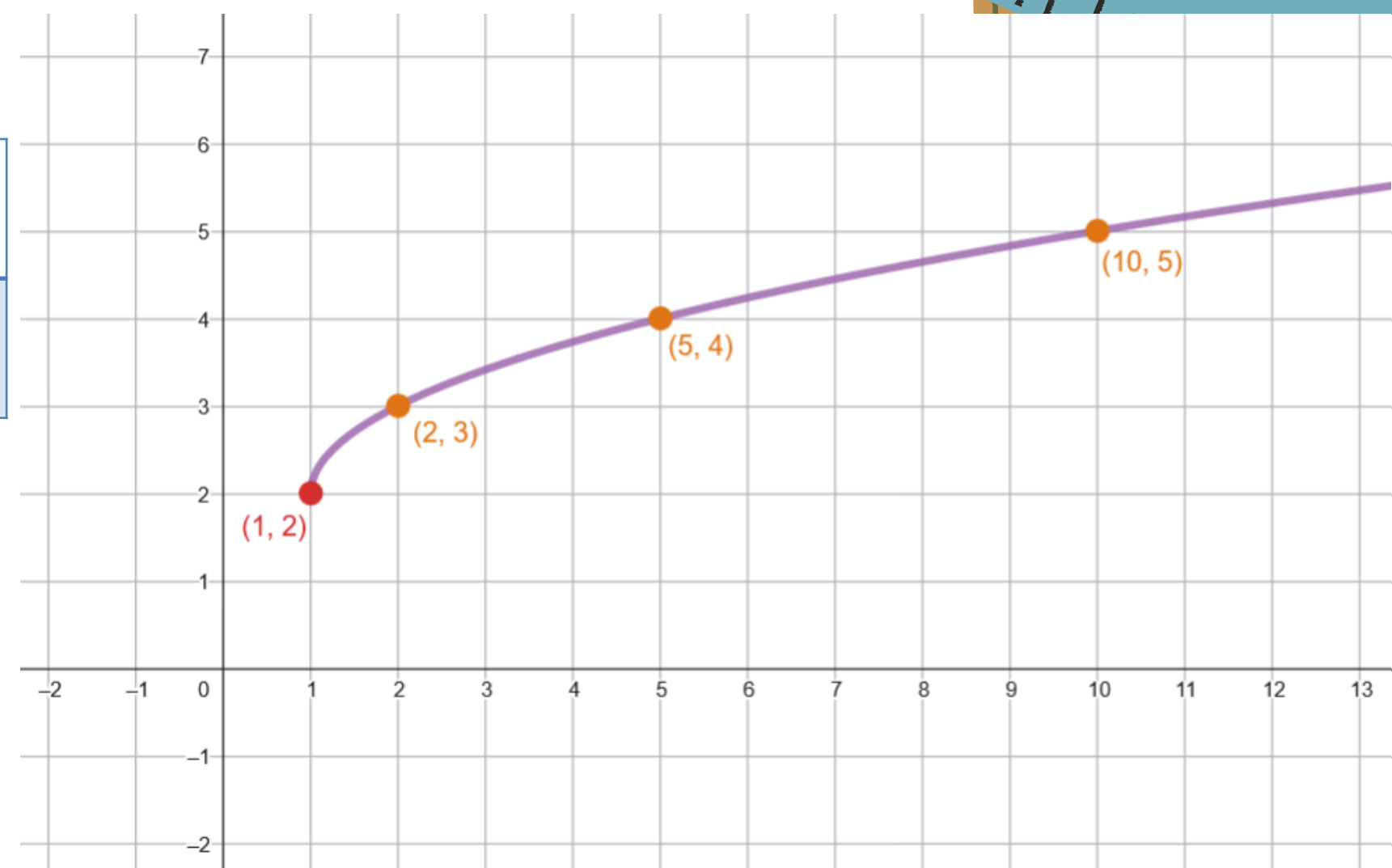
Domain: $x \geq h \Rightarrow x \geq 1$

Range: $a > 0 \Rightarrow y \geq 2$

Step 3: Construct Table of Values

x	1	2	5	10
y	2	3	4	5

Step 4: Plot the points



Square Root Function

Sketch the Graph: $f(x) = -\sqrt{x+2} + 4$

Step 1: Determine the starting point (h, k),
Starting point (-2, 4)

Step 2: Determine the Domain and the Range

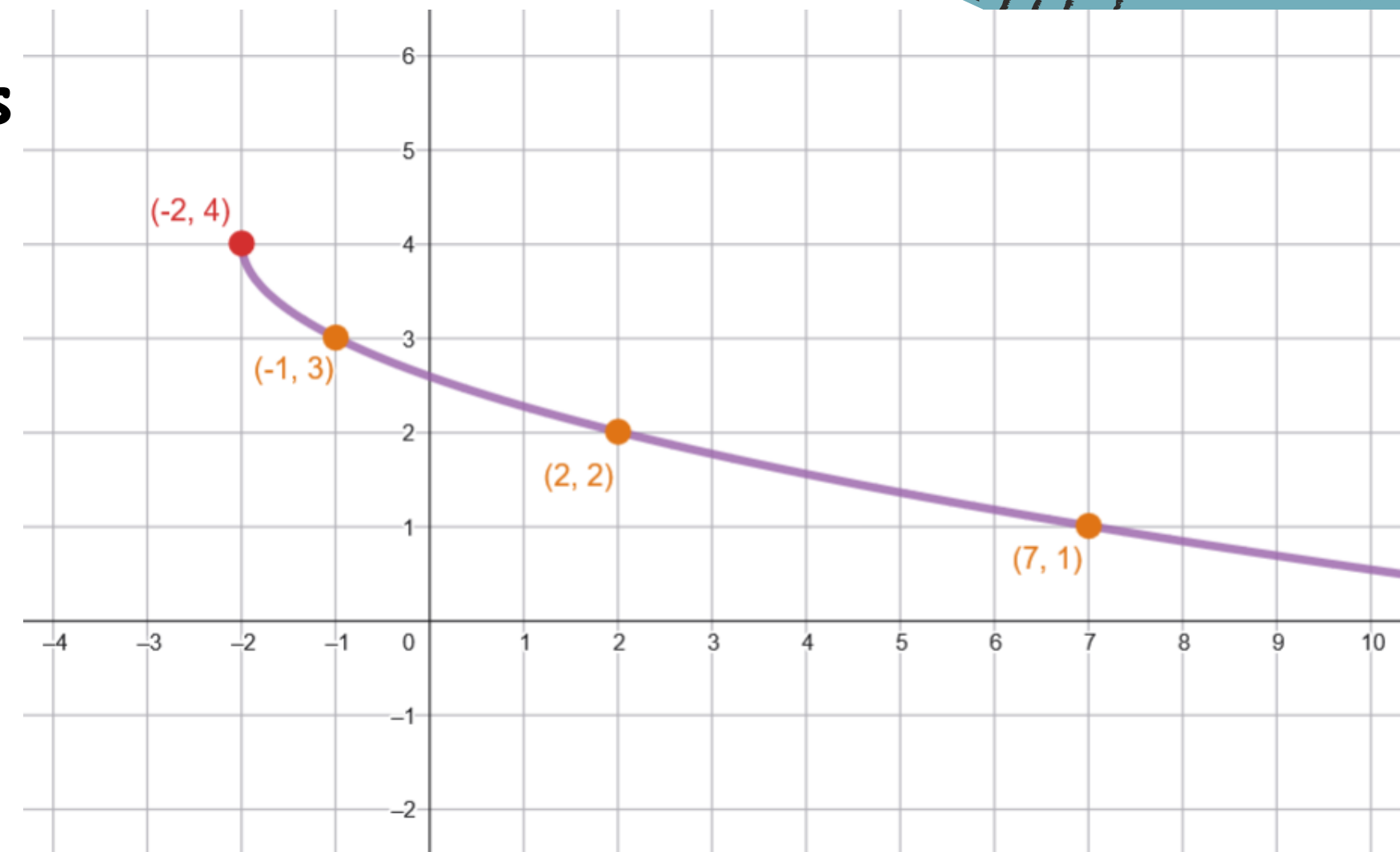
Domain: $x \geq h \Rightarrow x \geq -2$

Range: $a < 0 \Rightarrow y \leq 4$

Step 3: Construct Table of Values

x	-2	-1	2	7
y	4	3	2	1

Step 4: Plot the points



Square Root Function

Sketch the Graph: $f(x) = 2\sqrt{x-1} + 2$

Step 1: Determine the starting point (h, k),
Starting point (1, 2)

Step 2: Determine the Domain and the Range

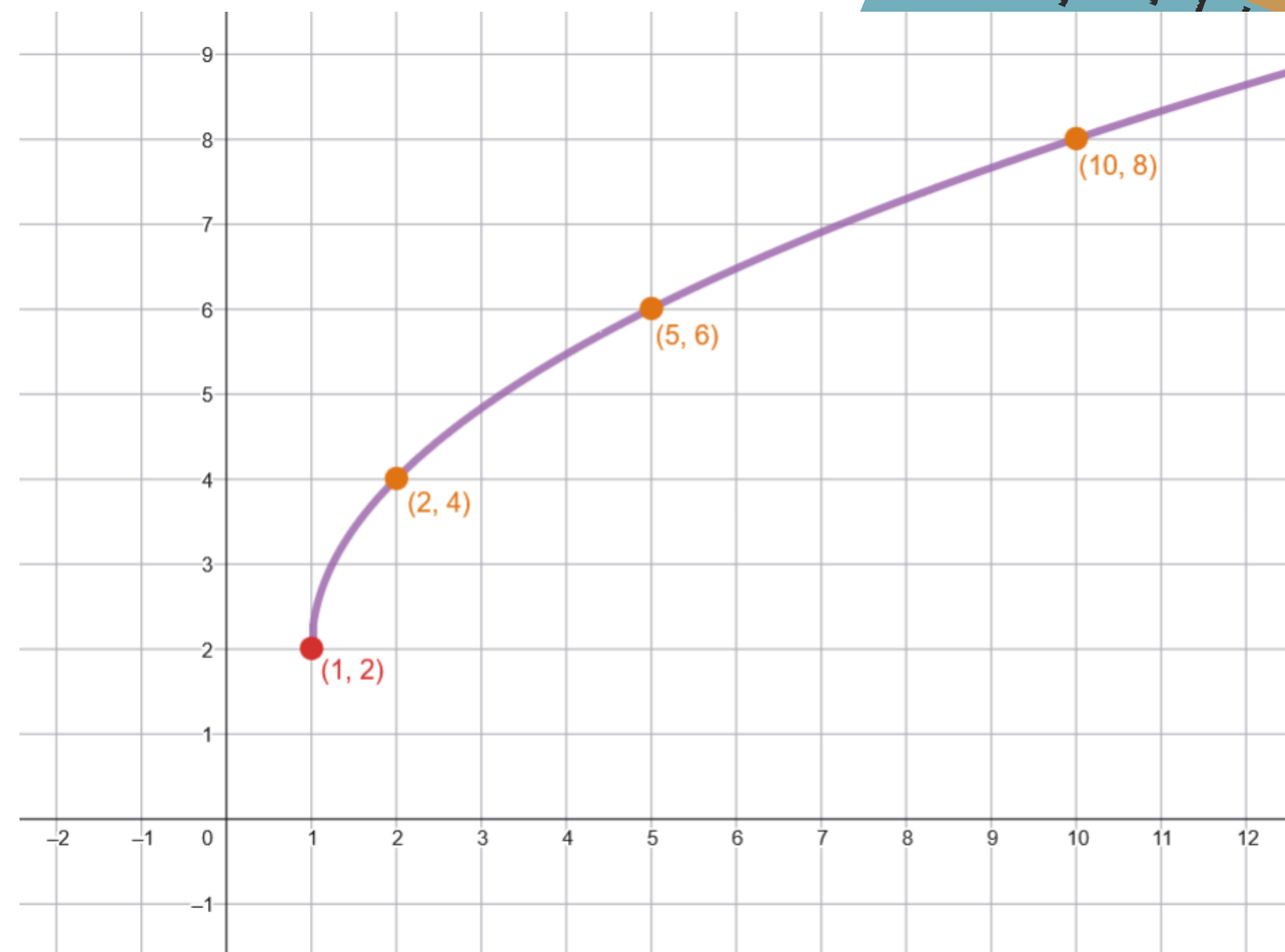
Domain: $x \geq h \Rightarrow x \geq 1$

Range: $a > 0 \Rightarrow y \geq 2$

Step 3: Construct Table of Values

x	1	2	5	10
y	2	4	6	8

Step 4: Plot the points



Key Characteristics

Cube Root Function

General form: $f(x) = a \cdot \sqrt[3]{(x - h)} + k$.

From the equation: The variable x appears underneath a radical sign with an index of 3 (a cube root).

From the graph: The graph forms a gentle S-shaped curve that passes through an inflection point and extends in both directions without restriction.

Domain and Range: Unlike the square root function, both the domain and range of a cube root function are all real numbers, since a negative number can have a real cube root.



Key Characteristics

Cube Root Function

Key Characteristics

- ❑ Domain: all real numbers.
- ❑ Range: all real numbers.
- ❑ Graph shape: an S-shaped curve that passes through the point (h, k) and is symmetric about that point (point symmetry).



Cube Root Function

Steps in Graphing

Step 1: Identify the point of symmetry (h, k),

- From the equation, extract (h) and (k)
- Example: $f(x) = \sqrt[3]{x+1} - 2 \rightarrow$ Point of Symmetry $((-1, -2))$

Step 2: Determine the Domain and the Range

Step 3: Construct Table of Values

Choose x-values on both sides of h that make the radicand a perfect cube (e.g., $\pm 1, \pm 8$) for easy computation.

Step 4: Plot the points

Plot the points and connect them with a smooth, continuous S-shaped curve.

Cube Root Function

Sketch the Graph: $f(x) = \sqrt[3]{x+1} - 2$

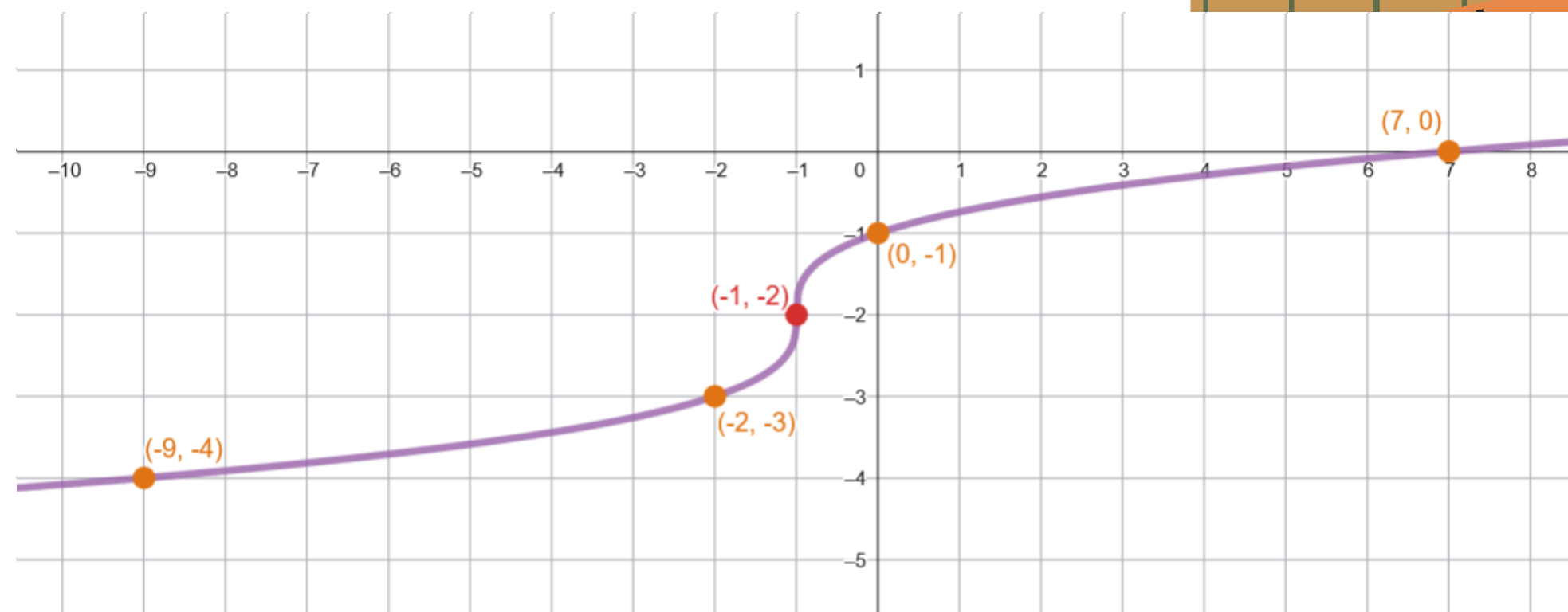
Step 1: Identify the point of symmetry (h, k),
Point of Symmetry: (-1, -2)

Step 2: Determine the Domain and the Range
Domain: all real numbers
Range: all real numbers

Step 3: Construct Table of Values

x	-9	-2	-1	0	7
y	-4	-3	-2	-1	0

Step 4: Plot the points



Cube Root Function

Sketch the Graph: $f(x) = -\sqrt[3]{x+2} + 4$

Step 1: Identify the point of symmetry (h, k),

Point of Symmetry: $(-2, 4)$

Step 2: Determine the Domain and the Range

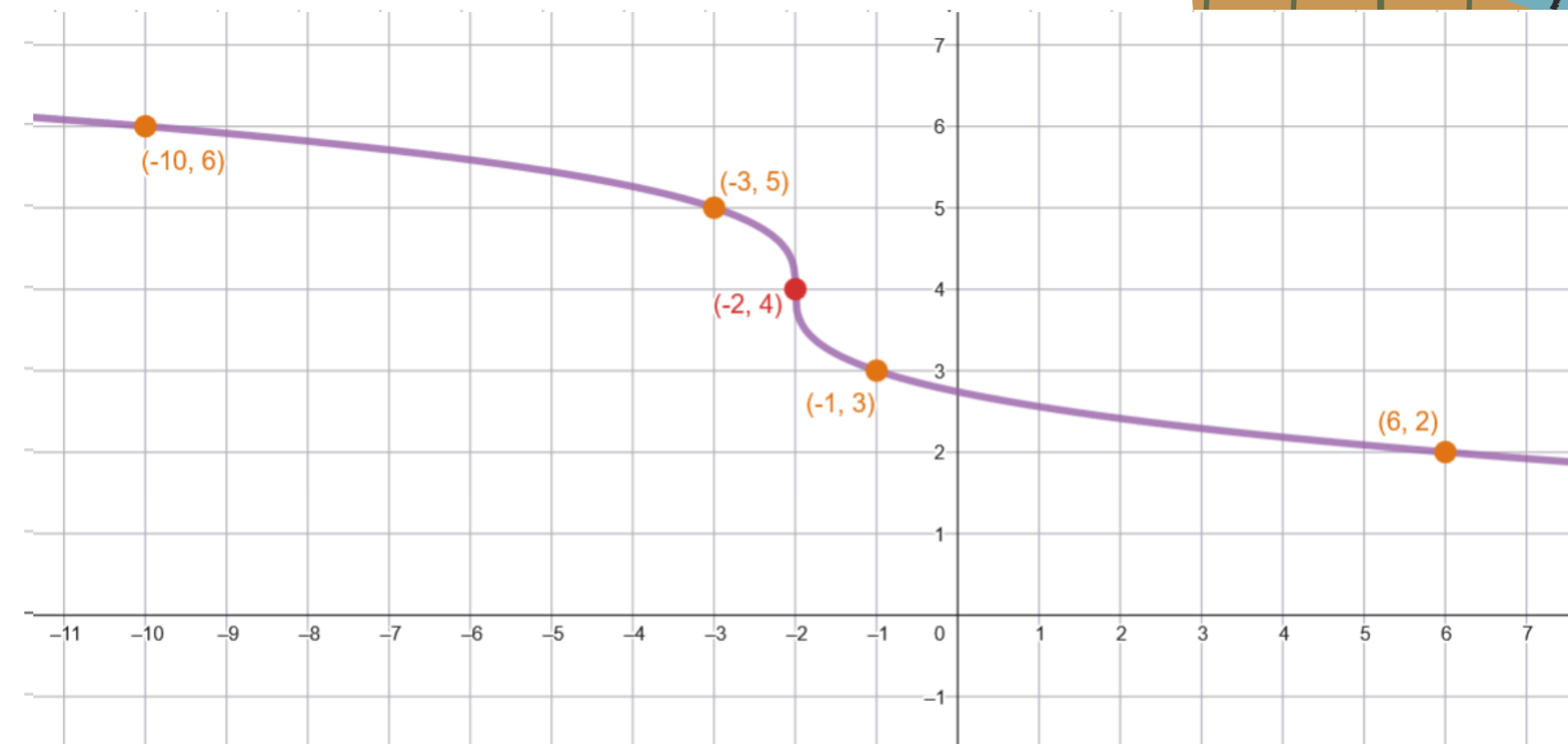
Domain: all real numbers

Range: all real numbers

Step 3: Construct Table of Values

x	-10	-3	-2	-1	6
y	6	5	4	3	2

Step 4: Plot the points



Cube Root Function

Sketch the Graph: $f(x) = 2\sqrt[3]{x-1} + 2$

Step 1: Identify the point of symmetry (h, k),

Point of Symmetry: (1, 2)

Step 2: Determine the Domain and the Range

Domain: all real numbers

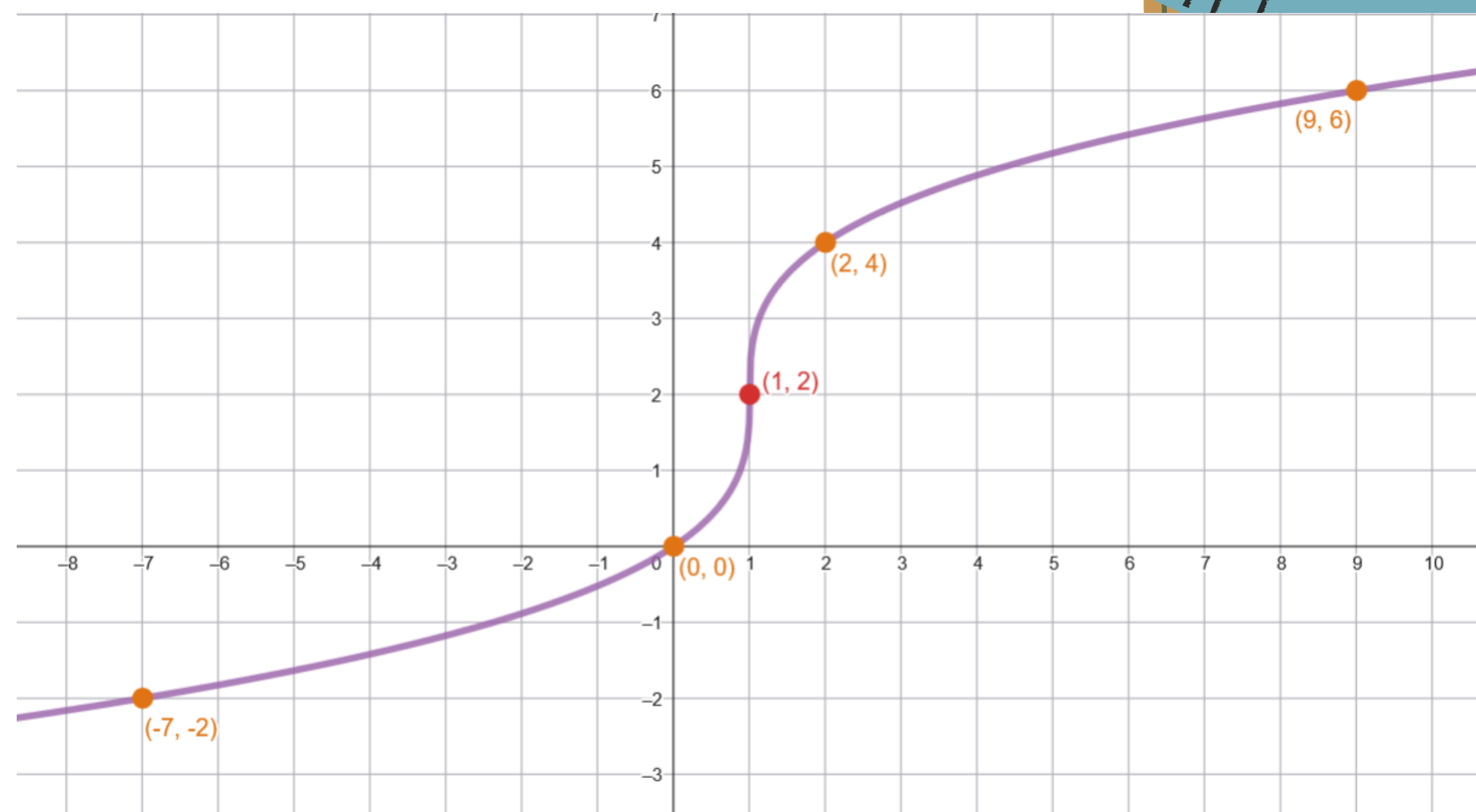
Range: all real numbers

Step 3:

Construct Table of Values

x	-7	0	1	2	9
y	-2	0	2	4	6

Step 4: Plot the points



Activity C.1: Finding Practical Application

Solve the following problems.

Problem 1: A household pays a fixed monthly connection fee of ₱300 and ₱10 per kilowatt-hour of electricity consumed.

- Write a linear function for the monthly bill B based on the number of kilowatt-hours k used.
- Calculate the bill if the household consumes 150 kilowatt-hours.

Solution:

Linear Function: $B(k) = 300 + 10k$

Total Bill:

Using the linear function, substitute $k = 150$

$$B(k) = 300 + 10k$$

$$B(150) = 300 + 10(150)$$

$$B(150) = 300 + 1500$$

$$B(k) = 1800$$

Answer:

The bill of the household is ₱1800.

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 2: You are selling tickets for fundraising event at a fixed price of ₱100 per ticket. The fixed cost to organize the event is ₱5,000.

- Write a linear function for the profit P based on the number of tickets x sold.
- Calculate the profit for selling 50, 100, 200, and 350 tickets.

Solution:

Linear Function: $P(x) = 100x - 5000$

Profit for selling 50 tickets:

$$P(50) = 100(50) - 5000$$

$$P(50) = 5000 - 5000$$

$$P(50) = 0$$

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 2: You are selling tickets for fundraising event at a fixed price of ₱100 per ticket. The fixed cost to organize the event is ₱5,000.

- Write a linear function for the profit P based on the number of tickets x sold.
- Calculate the profit for selling 50, 100, 200, and 350 tickets.

Solution:

Profit for selling 100 tickets:

$$P(x) = 100x - 5000$$

$$P(100) = 100(100) - 5000$$

$$P(100) = 10000 - 5000$$

$$P(100) = 5000$$

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 2: You are selling tickets for fundraising event at a fixed price of ₱100 per ticket. The fixed cost to organize the event is ₱5,000.

- Write a linear function for the profit P based on the number of tickets x sold.
- Calculate the profit for selling 50, 100, 200, and 350 tickets.

Solution:

Profit for selling 200 tickets:

$$P(x) = 100x - 5000$$

$$P(200) = 100(200) - 5000$$

$$P(200) = 20000 - 5000$$

$$P(200) = 15000$$

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 2: You are selling tickets for fundraising event at a fixed price of ₱100 per ticket. The fixed cost to organize the event is ₱5,000.

- Write a linear function for the profit P based on the number of tickets x sold.
- Calculate the profit for selling 50, 100, 200, and 350 tickets.

Solution:

Profit for selling 350 tickets:

$$P(x) = 100x - 5000$$

$$P(350) = 100(350) - 5000$$

$$P(350) = 35000 - 5000$$

$$P(350) = 30000$$

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 3: A ball is thrown into the air, and its height is modeled by $h(t) = -4.9t^2 + 20t + 1.5$, where t is the time in seconds, and $h(t)$ is the height in meters.

1. At what time does the ball reach its maximum height?
2. What is the maximum height of the ball?
3. When does the ball hit the ground?
4. Graph the function and explain the shape in terms of the ball's motion.

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 3: A ball is thrown into the air, and its height is modeled by $h(t) = -4.9t^2 + 20t + 1.5$, where t is the time in seconds, and $h(t)$ is the height in meters.

1. At what time does the ball reach its maximum height?

Solution:

The time of maximum height of the ball is at: $t = -\frac{b}{2a}$

$$a = -4.9$$

$$b = 20$$

$$c = 1.5$$

$$t = -\frac{20}{2(-4.9)} = -\frac{20}{-9.8} \approx \mathbf{2.04 \text{ seconds}}$$

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 3: A ball is thrown into the air, and its height is modeled by $h(t) = -4.9t^2 + 20t + 1.5$, where t is the time in seconds, and $h(t)$ is the height in meters.

2. What is the maximum height of the ball?

Solution:

The maximum height of the ball is solved by substituting $t = 2.04$ into $h(t)$.

$$h(t) = -4.9t^2 + 20t + 1.5$$

$$h(2.04) = -4.9(2.04)^2 + 20(2.04) + 1.5$$

$$h(2.04) = -20.39 + 40.8 + 1.5$$

$$h(2.04) \approx \mathbf{21.91 \text{ meters}}$$

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 3: A ball is thrown into the air, and its height is modeled by $h(t) = -4.9t^2 + 20t + 1.5$, where t is the time in seconds, and $h(t)$ is the height in meters.

3. When does the ball hit the ground?

Solution:

The ball hits the ground at $h(t) = 0$.

(Use quadratic formula to solve for t .) $\Leftarrow -4.9t^2 + 20t + 1.5 = 0$

$$a = -4.9 ; b = 20 ; c = 1.5$$

$$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$t = \frac{-(20) \pm \sqrt{(20)^2 - 4(-4.9)(1.5)}}{2(-4.9)} = \frac{-20 \pm \sqrt{400 + 29.4}}{-9.8} = \frac{-20 \pm \sqrt{429.4}}{-9.8}$$

$$t = \frac{-20 + \sqrt{429.4}}{-9.8} \approx -0.07 \quad t = \frac{-20 - \sqrt{429.4}}{-9.8} \approx 4.16$$

$$h(t) = -4.9t^2 + 20t + 1.5$$

$$0 = -4.9t^2 + 20t + 1.5$$

$$-4.9t^2 + 20t + 1.5 = 0$$

Answer:

The ball hits the ground at 4.16 seconds.

Activity C.1: Finding Practical Application

Solve the following problems.

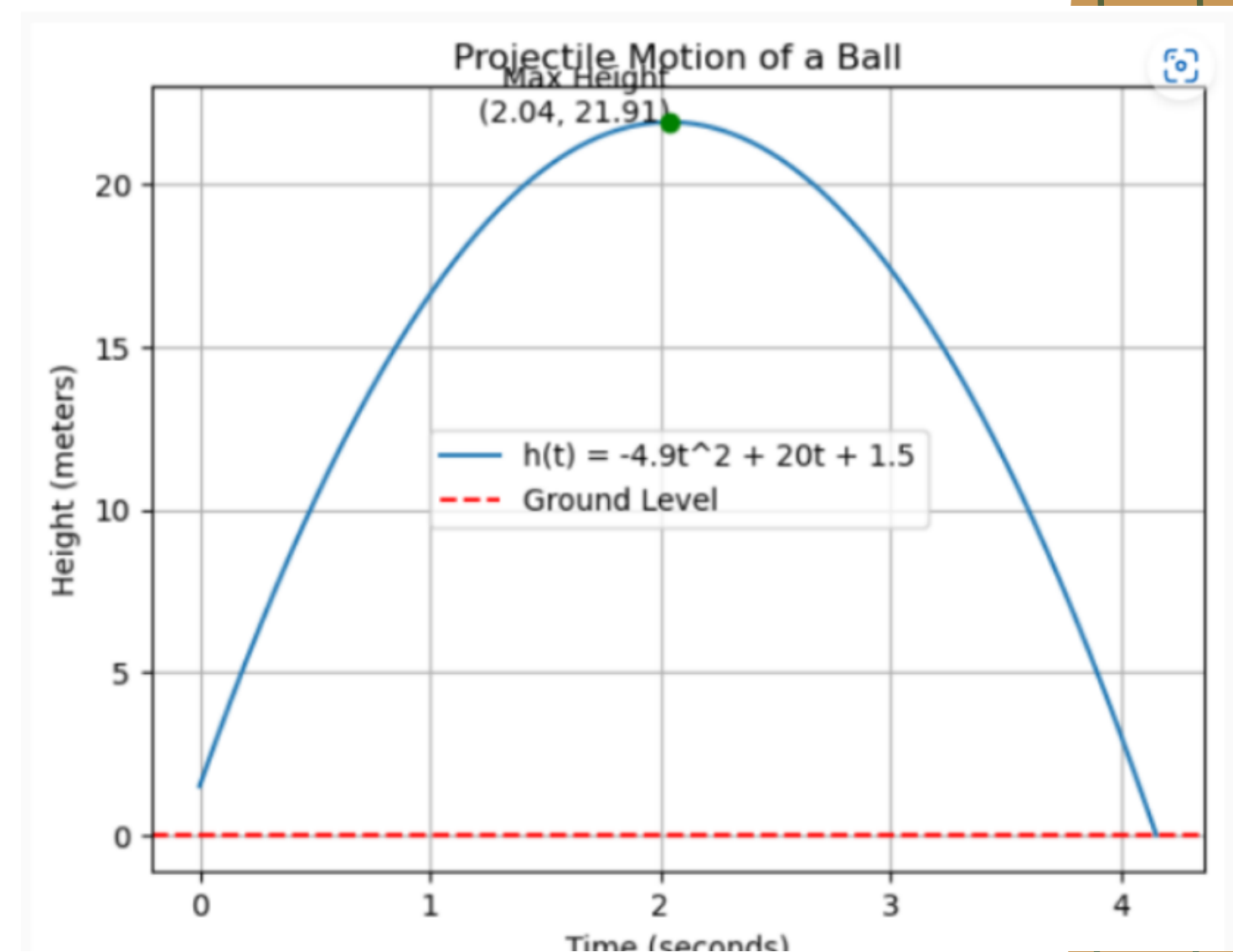
Problem 3: A ball is thrown into the air, and its height is modeled by $h(t) = -4.9t^2 + 20t + 1.5$, where t is the time in seconds, and $h(t)$ is the height in meters.

4. Graph the function and explain the shape in terms of the ball's motion.

Solution:

Vertex: (2.04, 21.91)

The graph is a parabola that opens downwards, representing the ball's motion. The vertex represents the maximum height, and the x -intercepts represent the times when the ball is at ground level.



Activity C.1: Finding Practical Application

Solve the following problems.

Problem 4: A small halo-halo stand in Burnham Park, Baguio City sells each cup for ₱40 and serves 120 customers per day. The owner is considering increasing the price. Based on customer feedback, for every ₱2 increase, they expect to lose 6 customers. Create model of the daily profit as a quadratic function of the number of ₱2 price increases.

Solution:

Step 1: Define the variables.

Let:

x = number of ₱2 price increases

$40 + 2x$ = new price per cup

$120 - 6x$ = number of customers

Profit = (price per cup) $\times$ (number of customers)

Activity C.1: Finding Practical Application

Solve the following problems.

Solution:

Step 2: Write the Profit Function.

$$P(x) = (40 + 2x)(120 - 6x)$$

Expand:

$$P(x) = 4800 - 240x + 240x - 12x^2$$

$$P(x) = -12x^2 + 4800$$

Step 3: Analyze the Function

This is a downward-opening parabola, and the maximum profit occurs at the vertex.

Since the linear term cancels out, the vertex is at $x = 0$.

So, the maximum profit is $P(0) = 4800$.

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 5: A farmer in Nueva Ecija wants to build a rectangular vegetable garden along the side of a barn, using 30 meters of fencing. Since one side is against the barn, she only needs to fence three sides (two widths and one length). What dimensions will maximize the area of the garden?

Solution:

Step 1: Define the variables.

Let:

x = length of each width (the two shorter sides)

$30 - 2x$ = the length

Step 2: Write the Area Function.

$$\text{Area} = \text{length} \times \text{width} = (30 - 2x)(x)$$

$$A(x) = -2x^2 + 30x$$

Activity C.1: Finding Practical Application

Solve the following problems.

Solution:

Step 3: Find the Maximum Area.

$$A(x) = -2x^2 + 30x$$

This is a quadratic function opening downward, so the maximum occurs at the vertex.

$$x = \frac{-b}{2a} = \frac{-30}{2(-2)} = 7.5$$

Substitute back to find the length:

$$\text{length} = 30 - 2x = 30 - 2(7.5) = 15$$

$$\text{Maximum Area} = 15 \times 7.5 = 112.5 \text{ m}^2$$

Final Answers: Width = 7.5 meters

Length = 15 meters

Maximum Area = 112.5 square meters

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 6: A factory produces metal rods that should be exactly 25 cm long. Due to machine variance, an inspector rejects any rod whose length differs from 25 cm by more than 0.5 cm. Model the deviation $D(x) = |x - 25|$ for a rod of length x , and determine whether a rod measuring 25.7 cm passes inspection.

Solution:

Step 1: Set up the model.

The deviation from the ideal length is represented by an absolute value function.

$$D(x) = |x - 25|$$

Activity C.1: Finding Practical Application

Solve the following problems.

Solution:

Step 2: Substitute $x = 25.7$ into the function.

$$D(25.7) = |25.7 - 25|$$

$$D(25.7) = |0.7|$$

$$D(25.7) = 0.7$$

Step 3: Compare the deviation to the allowed tolerance of 0.5 cm.

$0.7 > 0.5$, so the rod exceeds the allowed tolerance.

Answer:

The rod measuring 25.7 cm does NOT pass inspection, since its deviation of 0.7 cm exceeds the 0.5 cm tolerance.

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 7: A thermostat is set to maintain a room at 22°C . The deviation of the actual temperature from this target is modeled by $D(x) = |x - 22|$. Find the deviation when the actual room temperature is 25.5°C .

Solution:

Step 1: Identify the function and given value:

$$D(x) = |x - 22|, x = 25.5$$

Step 2: Substitute $x = 25.5$:

$$D(25.5) = |25.5 - 22|$$

$$D(25.5) = |3.5|$$

$$D(25.5) = 3.5$$

Answer:

The temperature deviates by 3.5°C from the target setting.

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 8: A landscaper needs to find the radius of a circular garden bed given its area. This is modeled by $r(A) = \sqrt{\frac{A}{\pi}}$, where A is the area in square meters. Using $\pi \approx 3.1416$, find the radius of a garden bed with an area of 154 square meters.

Solution:

Step 1: Identify the function and given value:

$$r(A) = \sqrt{\frac{A}{\pi}}, \quad A = 154$$

Step 2: Substitute $A = 154$ and $\pi \approx 3.1416$:

$$r(154) = \sqrt{\frac{154}{3.1416}} \approx 7$$

Answer:

The radius of the garden bed is 7 meters.

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 9: The distance a person can see to the horizon is modeled by $d(h) = \sqrt{13h}$, where h is the observer's height above sea level in meters and $d(h)$ is the distance in kilometers. Find the horizon distance for an observer at a height of 52 meters.

Solution:

Step 1: Identify the function and given value:

$$d(h) = \sqrt{13h}, \quad h = 52.$$

Step 2: Substitute $h = 52$:

$$d(52) = \sqrt{13(52)}$$

$$d(52) = \sqrt{676}$$

$$d(52) = 26$$

Answer:

The observer can see 26 kilometers to the horizon.

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 10: A farm cooperative uses a cube-shaped bin to store grain, with a volume of 2,197 cubic meters. The edge length is modeled by $s(V) = \sqrt[3]{V}$. Find the edge length of the bin.

Solution:

Step 1: Identify the function and given value:

$$s(V) = \sqrt[3]{V}, V = 2197.$$

Step 2: Substitute $V = 2197$:

$$s(2197) = \sqrt[3]{2197}$$

$$s(2197) = 13$$

Answer:

The edge length of the grain storage bin is 13 meters.

Activity C.1: Finding Practical Application

Solve the following problems.

Problem 11: Using a simplified estimation formula, engineers model the radius of a spherical water tank as $r(V) = \sqrt[3]{0.75V}$, where V is the volume in cubic meters. Find the radius of a tank with a volume of 288 cubic meters.

Solution:

Step 1: Identify the function and given value:

$$r(V) = \sqrt[3]{0.75V}, \quad V = 288.$$

Step 2: Substitute $V = 288$:

$$r(288) = \sqrt[3]{0.75(288)}$$

$$r(288) = \sqrt[3]{216}$$

$$r(288) = 6$$

Answer:

The radius of the spherical water tank is 6 meters.



**Thank you for
listening!**